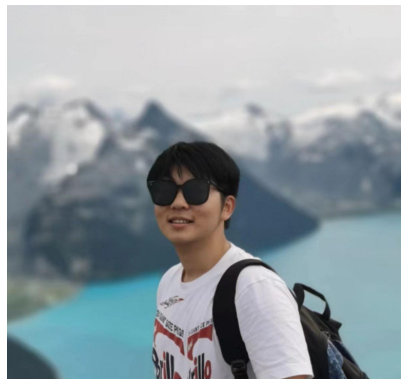
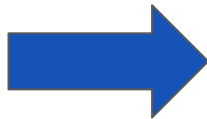


Welcome to the BabyVLM Tutorial!

Please visit our tutorial webpage for our schedule, slides, and data explorer!



Wenqi Wang



Max Whitton



Boqing Gong



Michael Frank



Shawn Wang

What are vision foundation models?

AI models pretrained on **large visual datasets** to support **many tasks**

General-purpose models

A new paradigm for computer vision and neuroscience research

Families of vision foundation models

Generative VLM

LLaVA / Qwen-VL



“What is happening?”



“They are high-fiving.”

Image generation

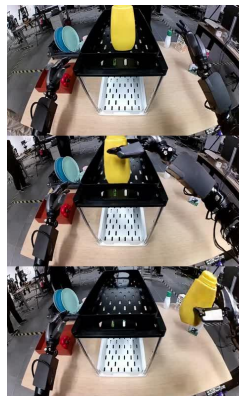
Stable Diffusion

“An astronaut playing piano in space”



World models

Cosmos-Predict2



Earlier

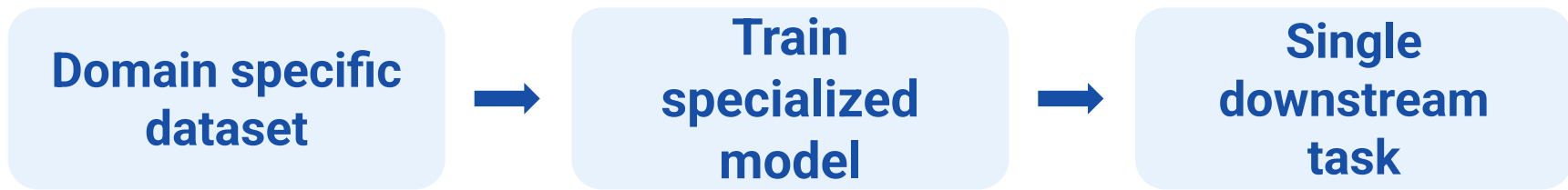


Later

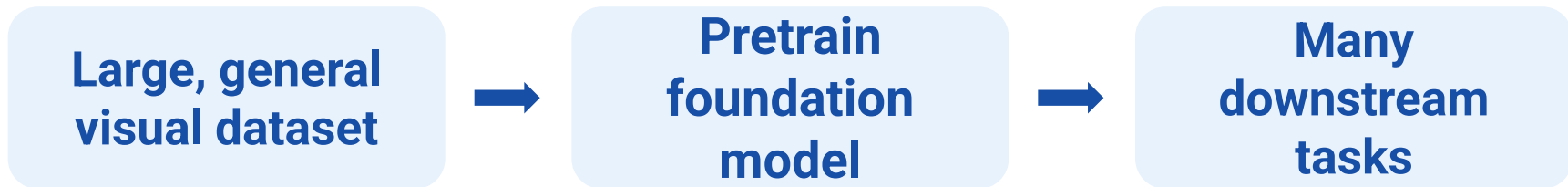
Observation + condition
→ possible future video

What's new with vision foundation models?

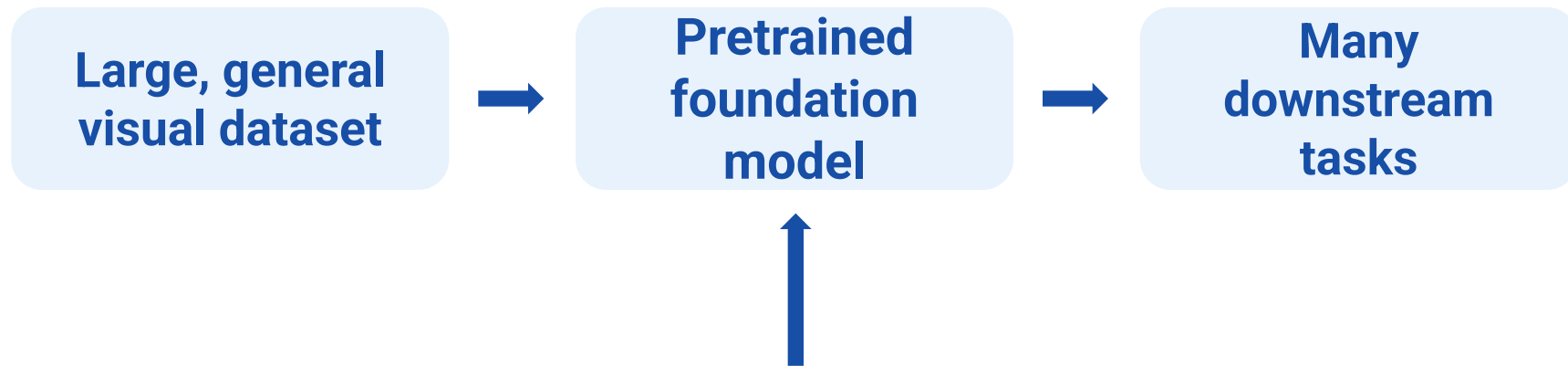
Old paradigm:



New paradigm:



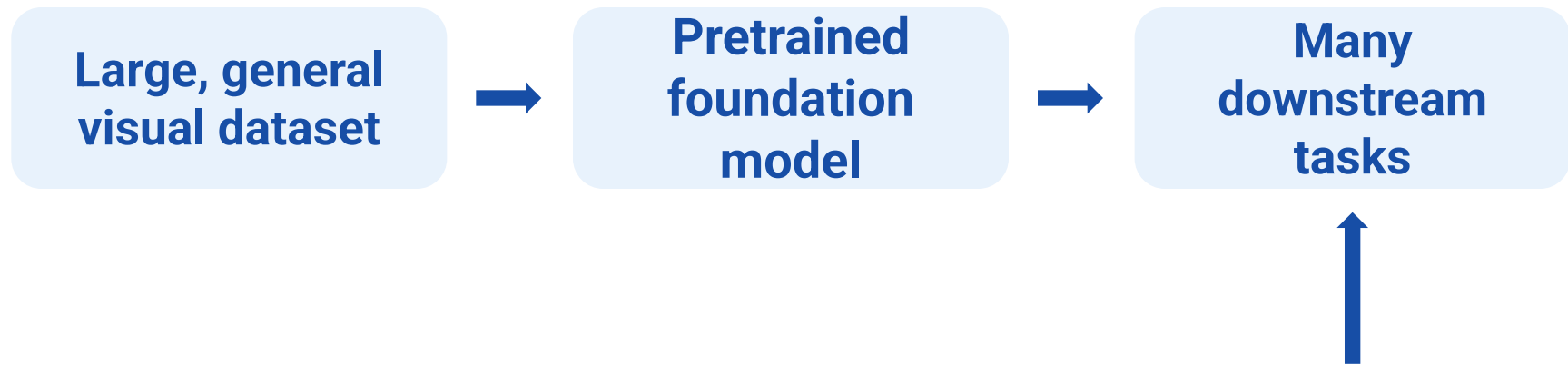
The problem with the new paradigm



This step is prohibitively expensive

Often concentrated in well-funded industry labs

The problem with the new paradigm



Researchers are forced to work here

Open source VFMs: **post-training**

Closed source VFMs: **"API research"**

Progress needs a broader community

Public trust

Academic pretraining lets us test how models learn from scratch.

Understanding this makes scrutiny possible.

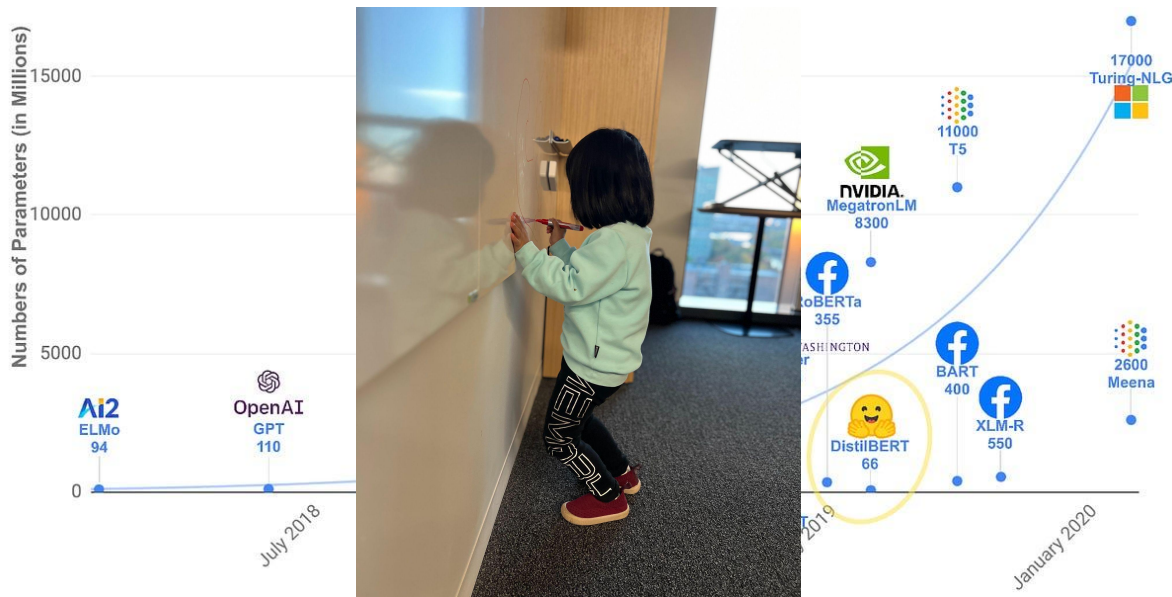
New ideas

A wider research community can explore what comes next.



Two proven paths to learning in foundation models

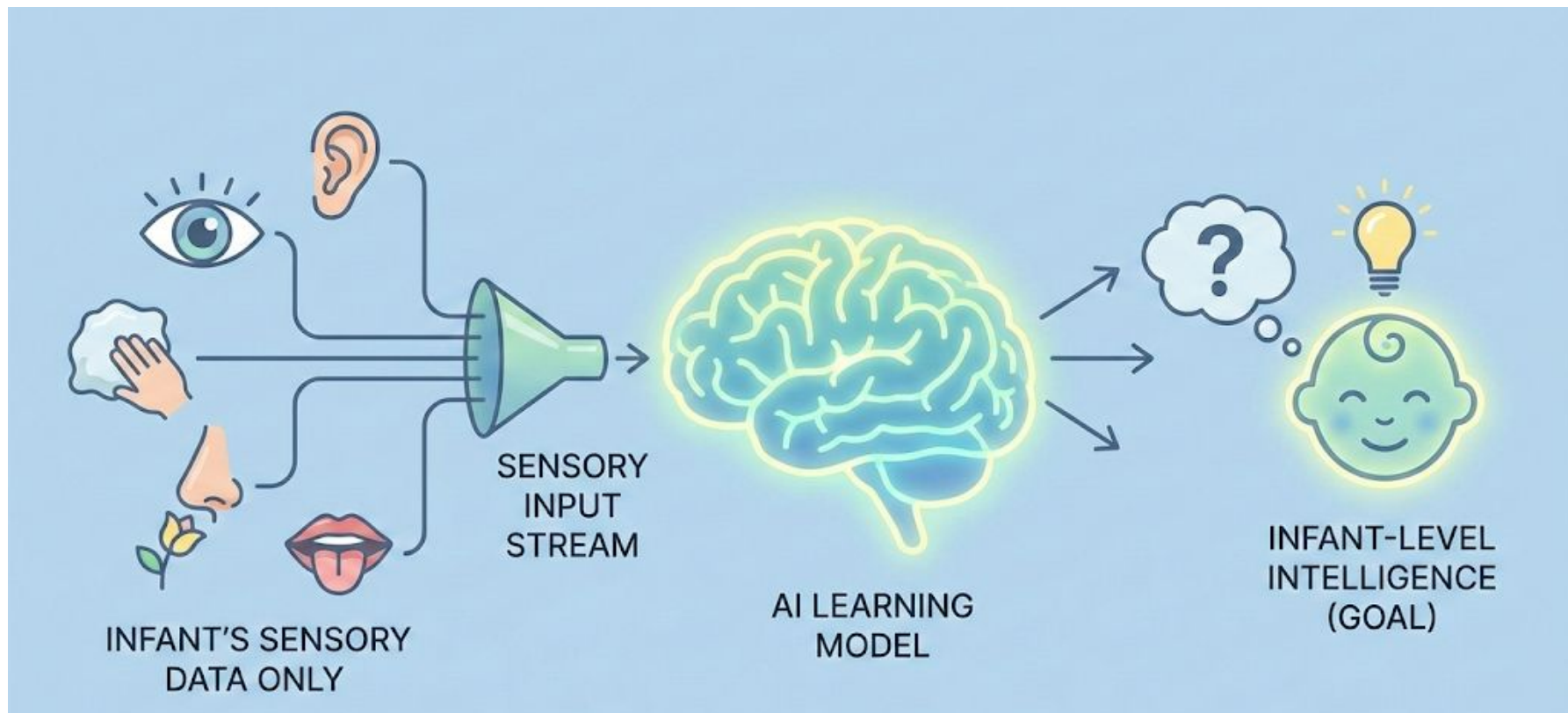
Human Scaling Development



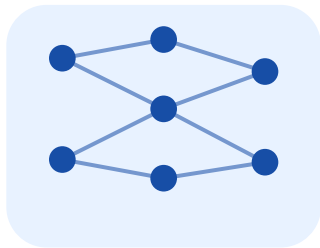
Artificial 3-year-old intelligence



Can we achieve infant-level intelligence using infants' sensory data?

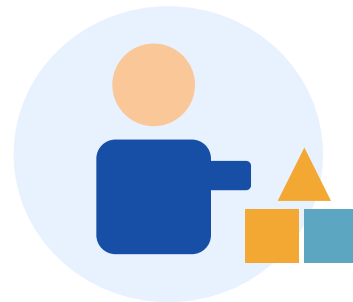


A bridge between vision and human development



Computer vision

Leverage existing learning theory to improve models.



Developmental psychology

Test ideas about learning from sensory experience.



Can we achieve infant-level intelligence using infants' sensory data?

Why should we care?

Scientifically reasonable

Computationally friendly

Safety, novelty, benefiting AGI, ...

Intriguing: Babies!



VideoPrism video corpus:
>100 years total duration



A three-year-old human:
< 3 years of total wake time



Can we achieve infant-level intelligence using infants' sensory data?

SAYCam: A Large, Longitudinal Audiovisual Dataset Recorded From the Infant's Perspective

Jessica Sullivan¹, Michelle Mei¹, Andrew Perfors², Erica Wojcik¹, and Michael C. Frank³



The BabyView dataset: High-resolution egocentric videos of infants' young children's everyday experiences

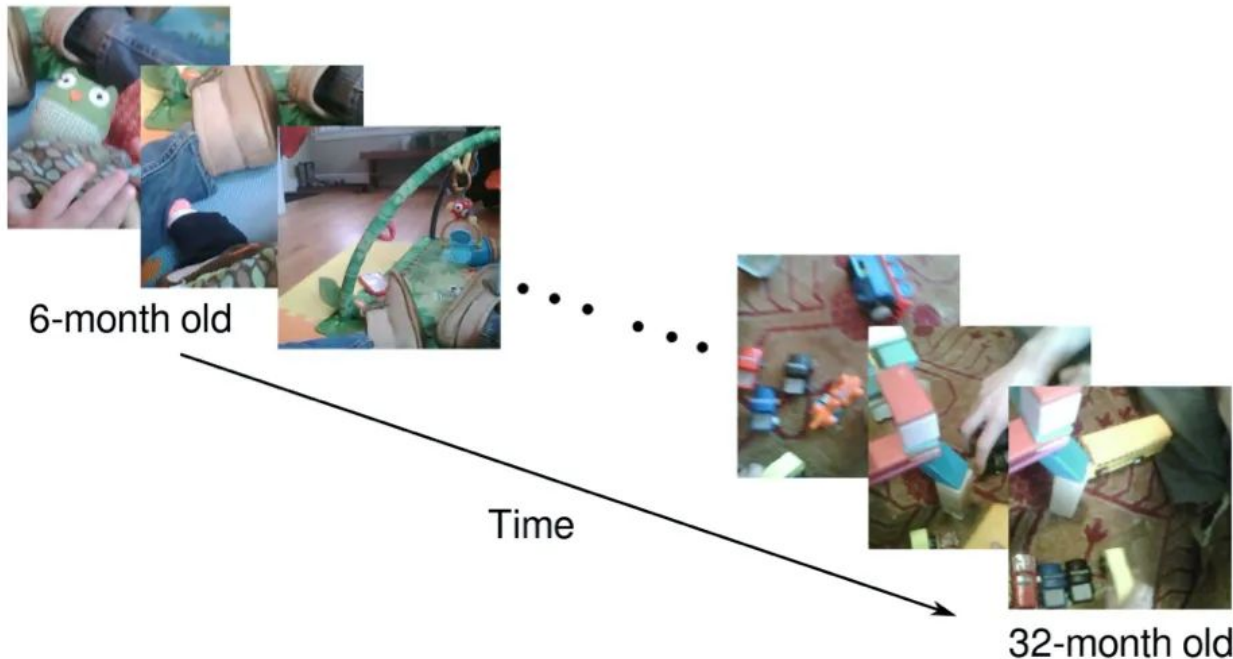


Bria Long^{1,2*} Robert Z. Sparks^{1*} Violet Xiang^{1*} Stefan Stojanov^{1*} Zi Yin¹
Grace E. Keene¹ Alvin W. M. Tan¹ Steven Y. Feng¹ Auddithio Nag¹ Chengxu Zhuang¹
Virginia A. Marchman¹ Daniel L. K. Yamins¹ Michael C. Frank¹



Can we achieve infant-level intelligence using infants' sensory data?

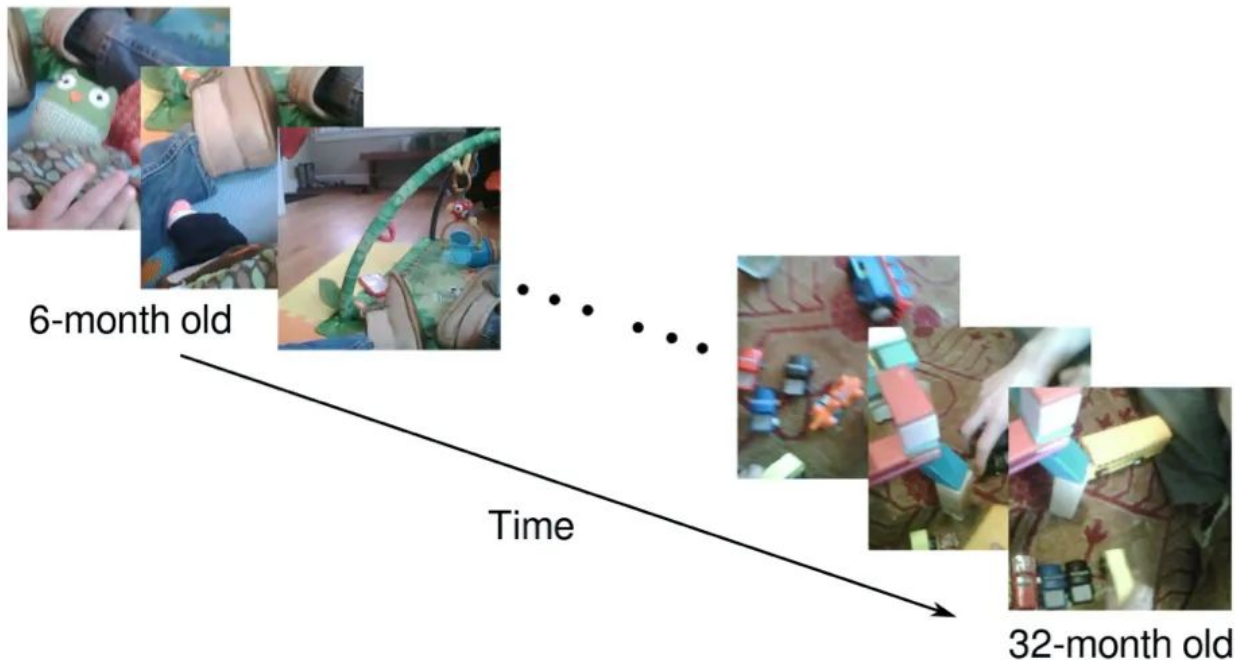
Longitudinal





Can we achieve infant-level intelligence using infants' sensory data?

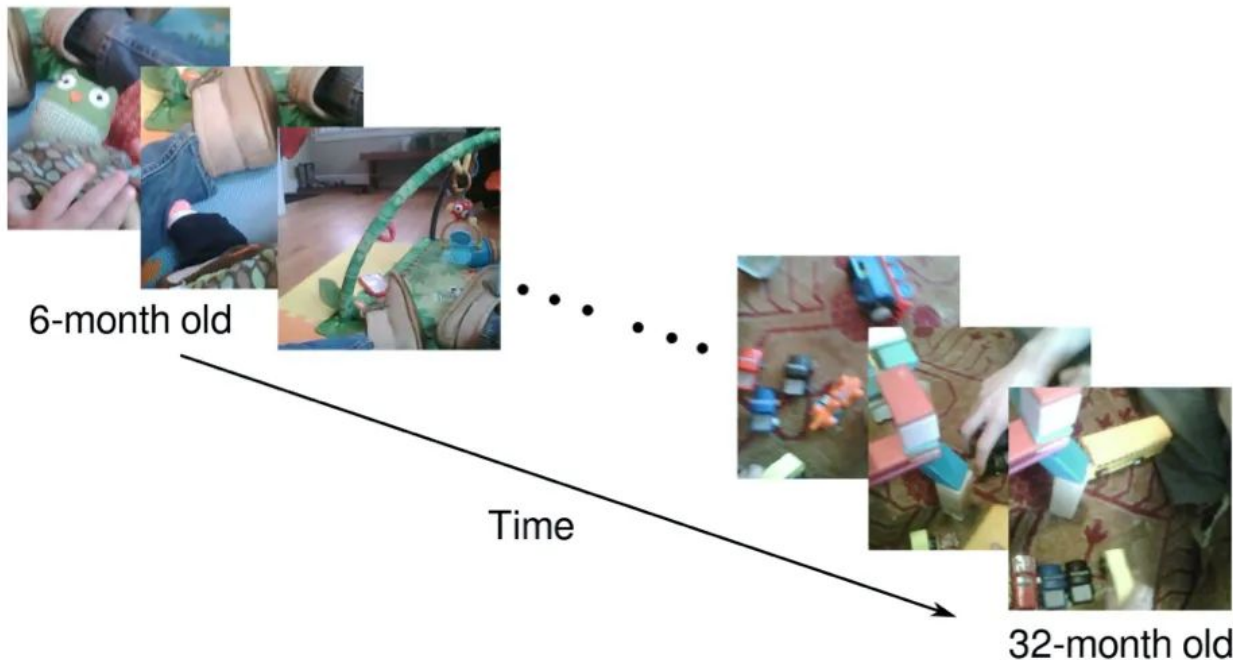
Longitudinal, egocentric





Can we achieve infant-level intelligence using infants' sensory data?

Longitudinal, egocentric video recording of what infants see and hear



What is a vision foundation model?

Different model families in different use cases

Generative VLM

LLaVA / Qwen-VL



“What is happening?”



“They are high-fiving.”

Image generation

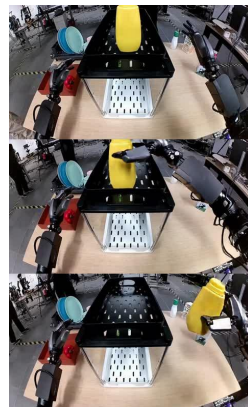
Stable Diffusion

“An astronaut playing piano in space”



World models

Cosmos-Predict2



Earlier



Later

Observation + condition
→ possible future video

Running example: generative vision-language models

Images and instructions in. Text out.

Image + instruction



"Describe this image."



**Vision-language
model (VLM)**

e.g., LLaVA or Qwen-VL

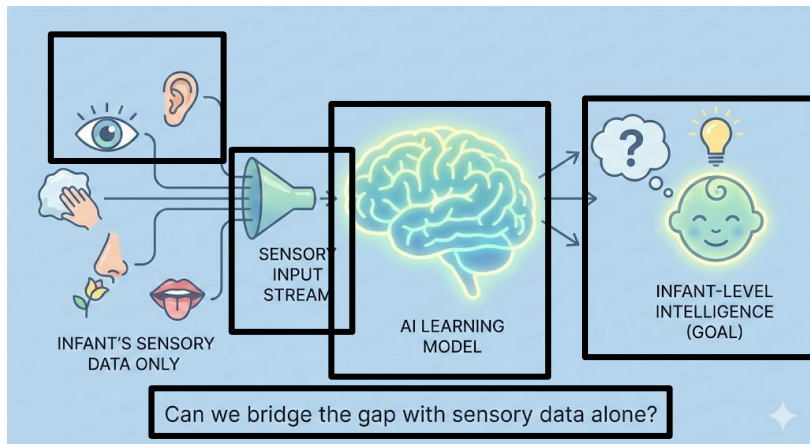


Text response

"A dog and a woman
are giving each other
a high-five on a beach."

Describe scenes, answer questions and read text.
Increasingly, these models also tackle visual reasoning.

Tutorial overview



1 Developmental data collection

***coffee break**

2 Pretraining data curation

3 Modeling and algorithm

4 Developmentally aligned benchmarks

5 Instruction tuning, evaluation and results

6 Future work and Q&A

Break for talk from Michael Frank

10:00-11:00

<https://meet.google.com/cjr-jkjd-rpd>

Coffee break

11:00-11:20

BabyVLM-v2: Pretraining Dataset

Toward **Developmentally Plausible** Vision Foundation Models



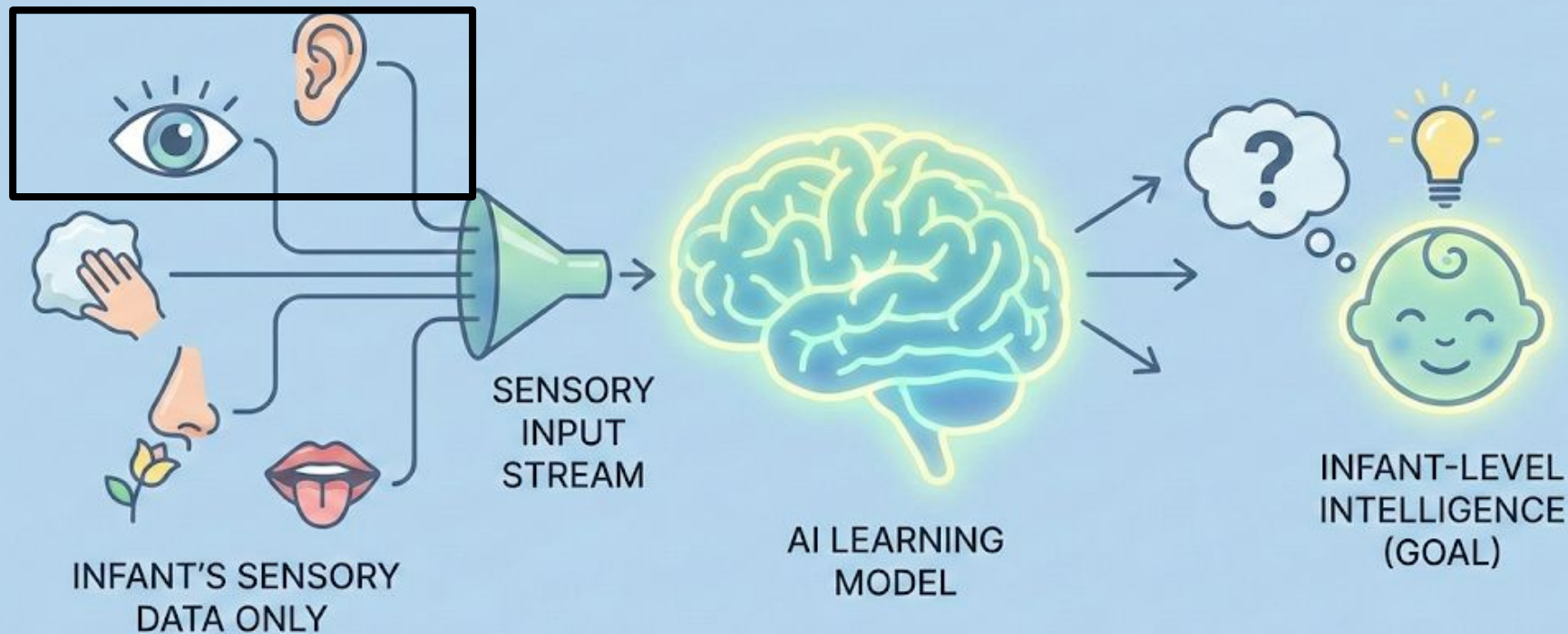
BabyVLM-V2

Wenqi Wang

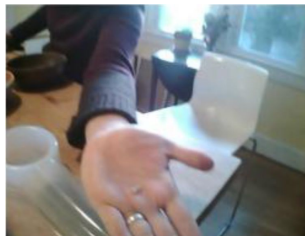
wqwang@bu.edu

Can we achieve infant level intelligence using infant's sensory data?

How to process raw sensory data?



SAYCam dataset



What is the recording frequency?

What objects do babies see?

Where do babies stay?

What do babies hear?

Recorded experience

3 infants

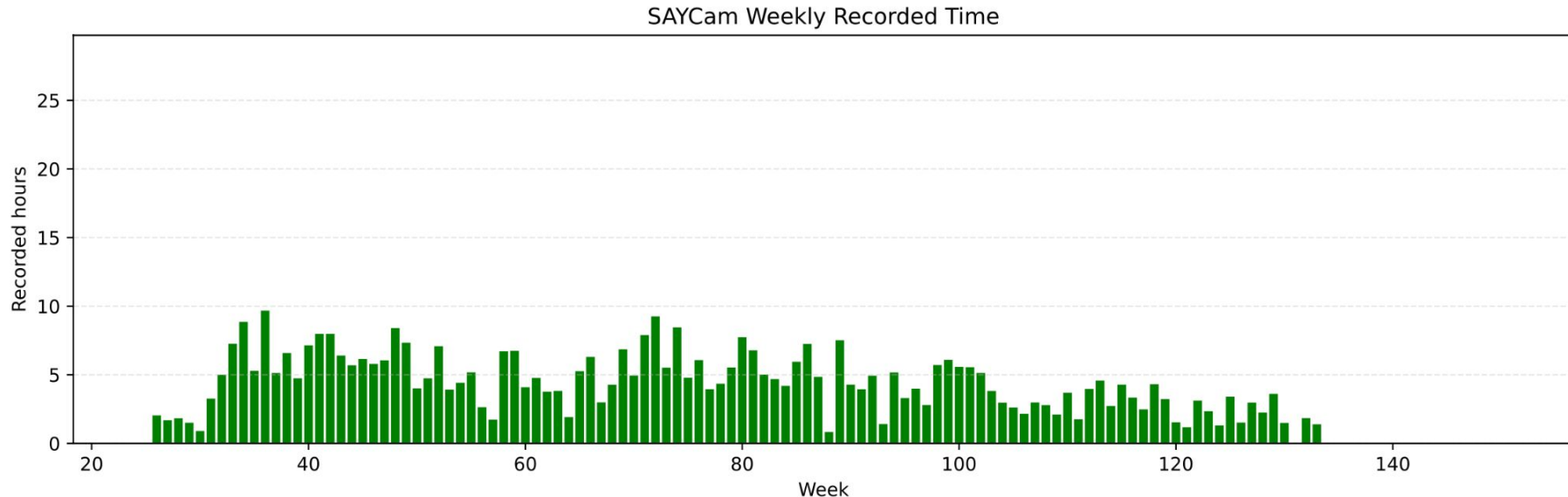
478 hours

~6–32 months

SAYCam dataset - statistical analysis

Recording frequency:

- ~4.5 hours / week
- distribute smoothly from 6 months - 32 months



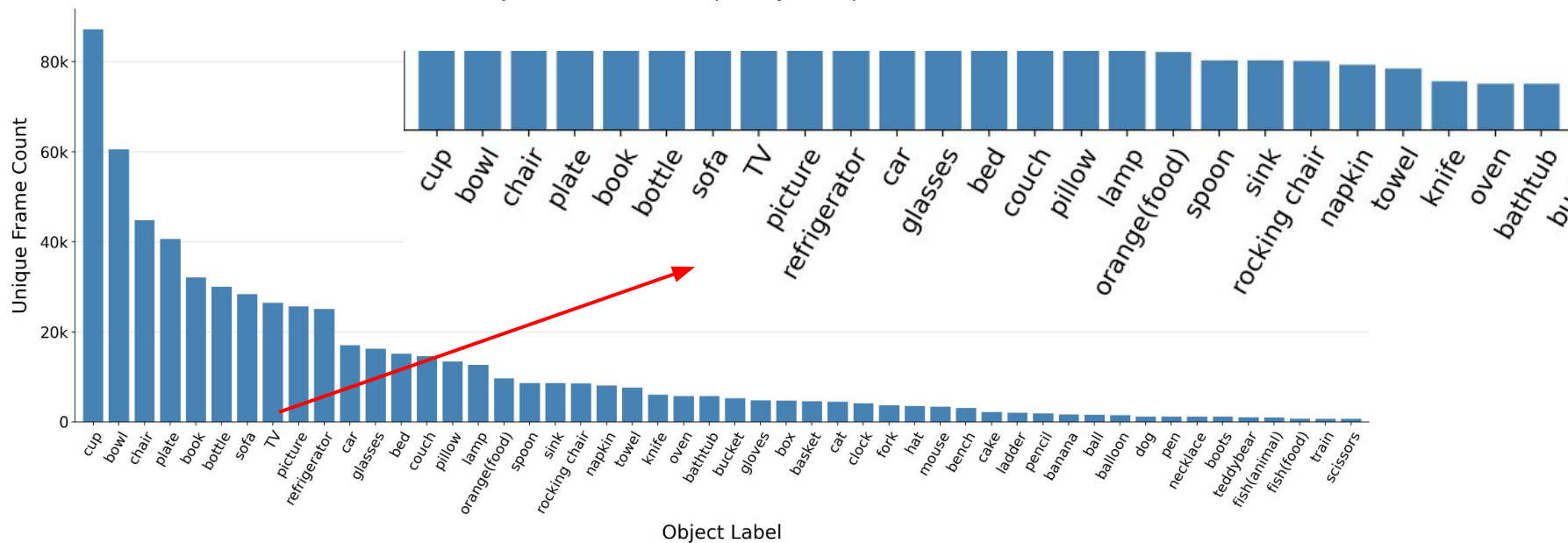
SAYCam dataset - statistical analysis

What do babies see and stay? Object detection

- Detection: 1 fps using YOLO8x-Worldv2
- Vocabulary: nouns from MB-CDI world bank (N=267)
- confidence > 0.65



SAYCam — Object Detection Frequency — Top 50 (threshold=0.65)



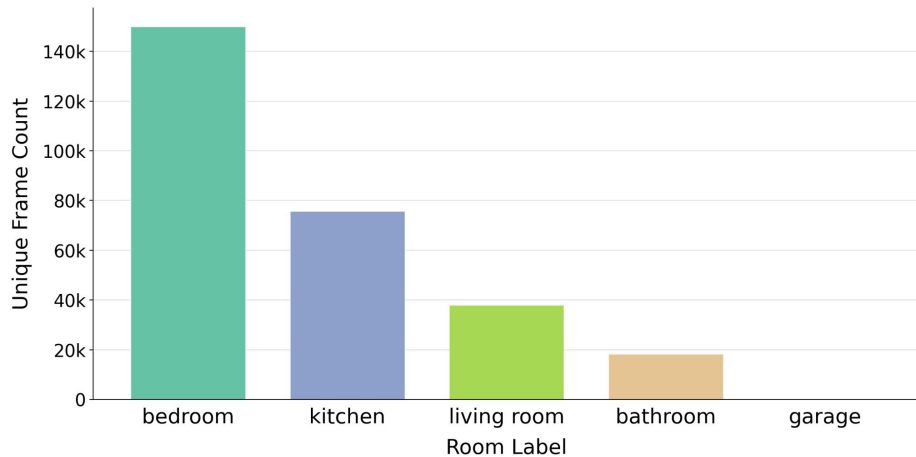
SAYCam dataset - statistical analysis

What do babies see and stay? Object detection

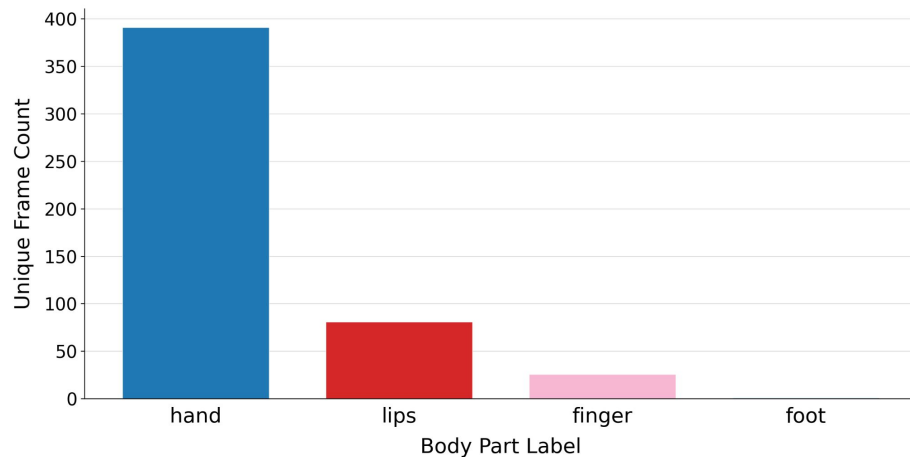
- Detection: 1 fps using YOLO8x-Worldv2
- Vocabulary: nouns from MB-CDI world bank (N=267)
- confidence > 0.65



SAYCam — Rooms at Home — Detection Frequency (threshold=0.65)



SAYCam — Body Parts — Detection Frequency (threshold=0.65)

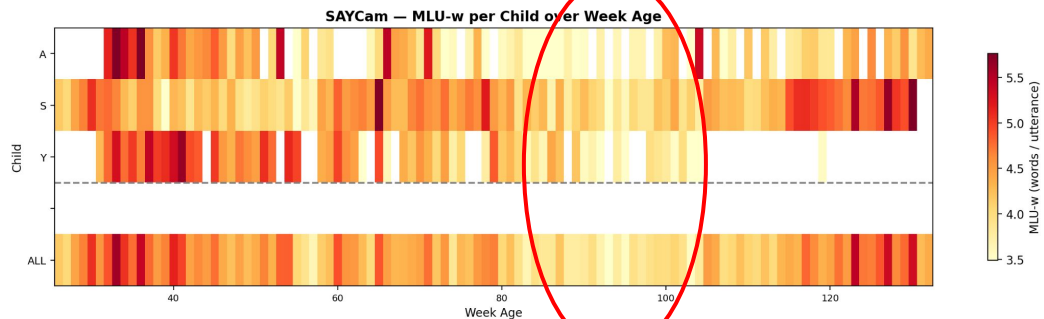
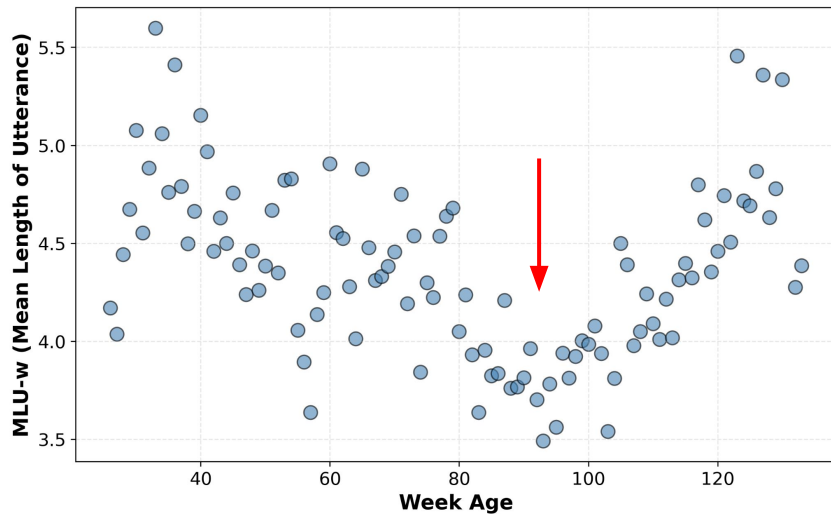


SAYCam dataset - statistical analysis

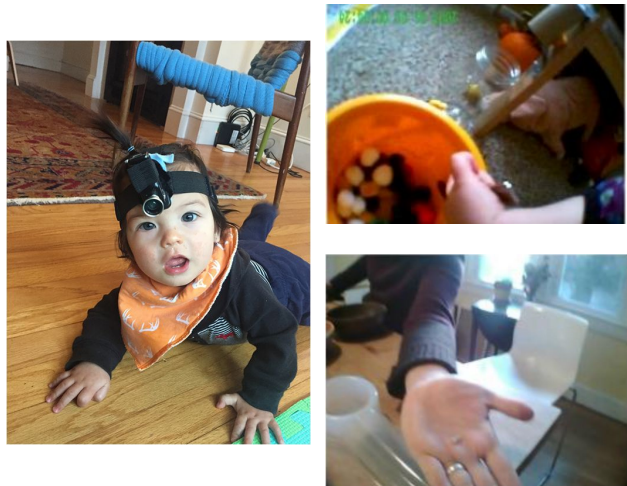
What do babies hear? Utterance analysis

- Transcription: Azure Speech Recognition
- MLU (mean length of utterance): 4.37 words overall

SAYCAM — MLU-w (Mean Length of Utterance) vs Week Age



From infant experience to model-ready data

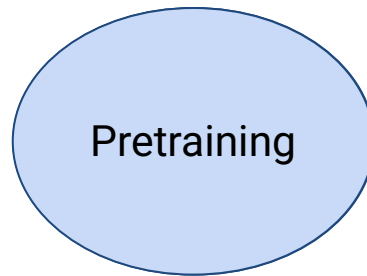
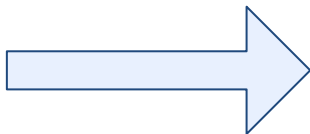


Recorded experience

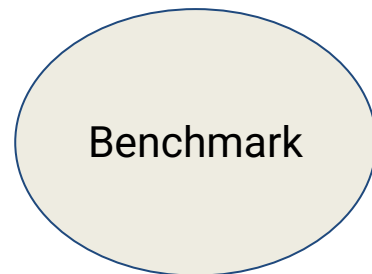
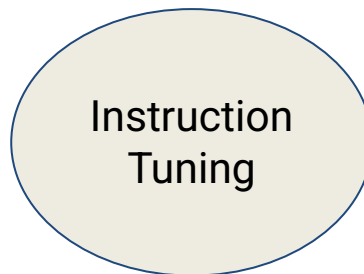
3 infants

478 hours

~6–32 months



Model-ready data



Pretraining data - two signals from the recording

Visual stream



Speech → text

Azure Speech Recognition

Utterance text

Start / end boundaries

Transcription confidence

			
Time: 15:22 Utterance: It's really good with yogurt.	Time: 17:03 Utterance: Want to try a little piece of this?	Time: 18:46 Utterance: How about another blueberry?	Time: 18:59 Utterance: Oh that's a very sweet one.

Pretraining data - split source clips

Split source clips first; derived training examples inherit the source split.

Raw SAYCam clips

Approximate source ratio 3:1:1



Pretraining source

Validation

Test

All clips containing speech

Remaining clips split equally

Pretraining + instruction tuning

**Task
validation**

**Task
testing**

Pretraining data - filter for usable alignment

Filter paired pretraining examples using threshold control

1 Segment speech

One transcript-defined clip per utterance

Duration ≥ 0.5 s

ASR confidence ≥ 0.3

2 Align video + text

X-CLIP similarity

> 0.1

Pad 1s on each side

3 Align frame + text

Sample clips at 1 FPS

CLIP similarity

> 0.2

Pretraining data - three formats

Video-utterance

P1: Video-Utterance Pair



Let's read the first book of sushi
by Amy Wilson Sanger



Image sequence + one utterance

Image-utterance

P2: Image-Utterance Pair



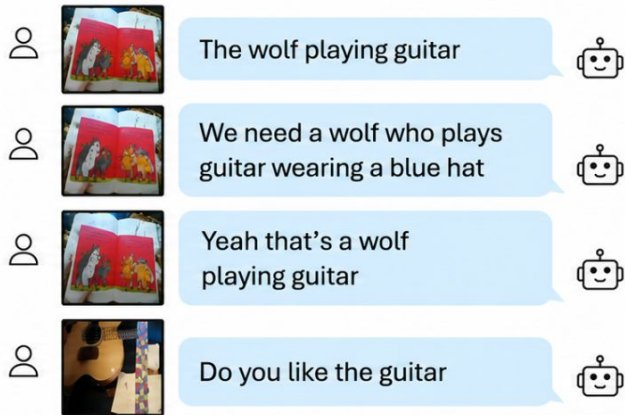
That's the baby dressed as a grape



One frame + one utterance

Interleaved sequence

P3: Multi-turn Image-Utterance



4-8 image-utterance pairs

The resulting pretraining data

Unimodal pools

No alignment filtering

~1.1M

image-only samples

283K

utterance-only
samples

Aligned multimodal formats

768K

image–utterance pairs

181K

video–utterance pairs

63K

interleaved sequences

BabyVLM-v2: Modeling and Algorithm

Toward **Developmentally Plausible** Vision Foundation Models



BabyVLM-V2

Shengao Wang

wsashawn@bu.edu

Presented by

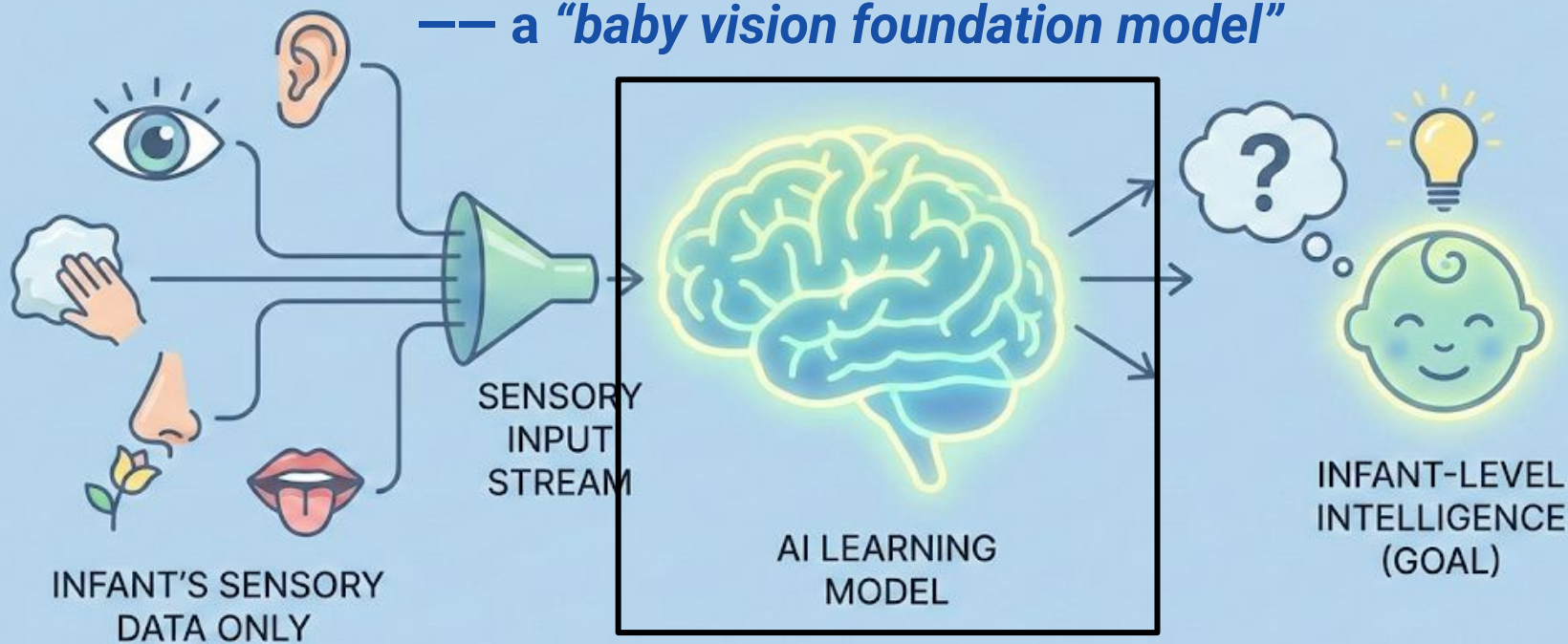
Max Whitton: maxwh@bu.edu



Can we achieve infant-level intelligence using infants' sensory data?

How to model “infant-level intelligence”?

— a “baby vision foundation model”



Our focus: generative vision-language models

Image + instruction



“Describe this image.”



**Vision-language
model (VLM)**

e.g., LLaVA or Qwen-VL



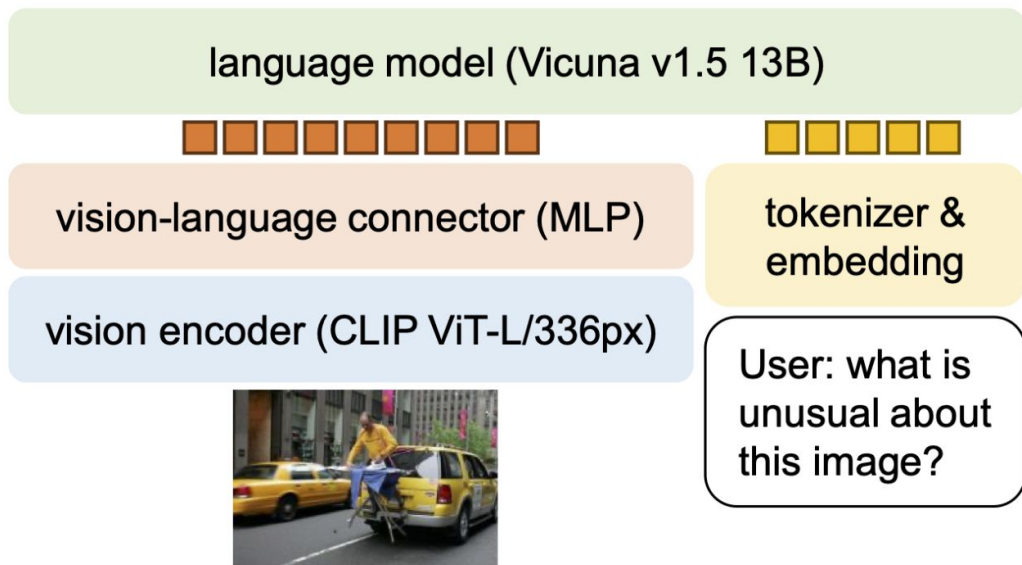
Text response

“A dog and a woman
are giving each other
a high-five on a beach.”

What is a generative vision foundation model?

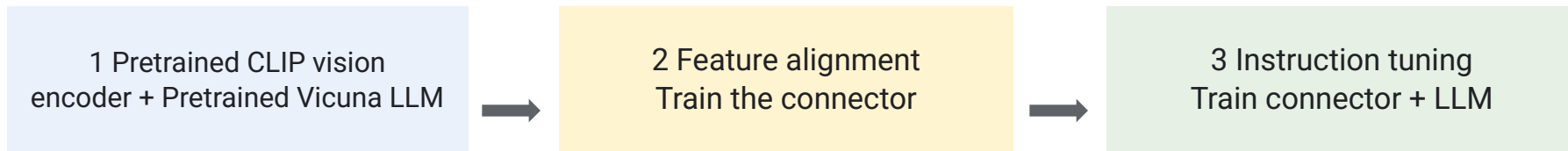
A generative vision foundation model usually comes with three components:

Vision backbone, language backbone, modality connector



How are vision-language models trained?

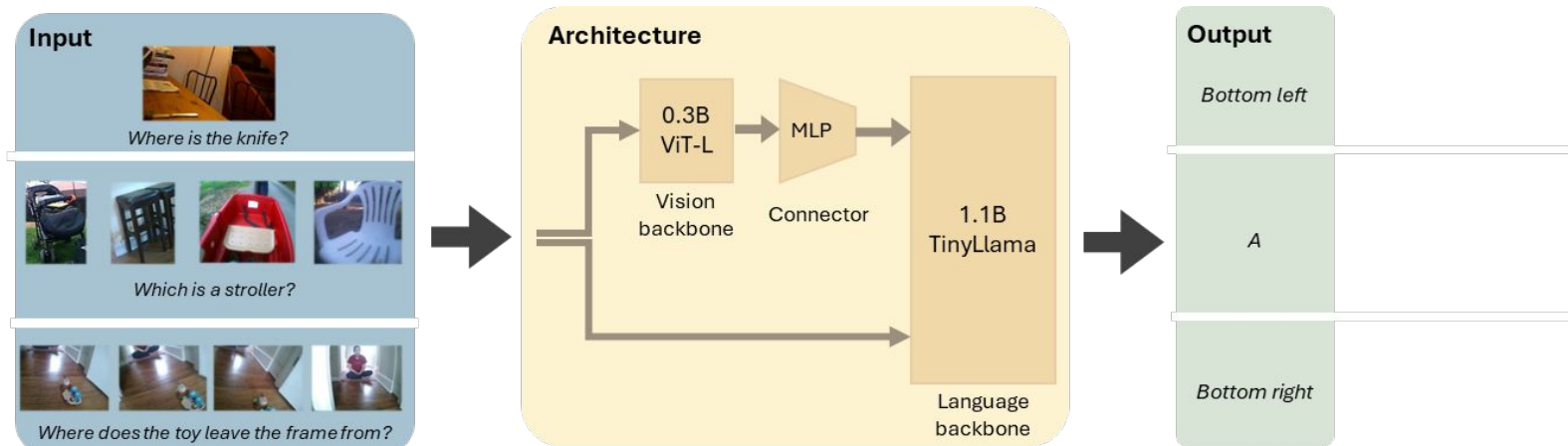
LLaVA: a concrete example



LLaVA-RLHF adds preference training.

Baseline model – Baby model

- **A compact, versatile Vision Language Model (VLM)**
 - Versatile input format: Single image, multiple images, video, multi-turn
 - Follow natural language instruction
- Trained from scratch on our pretrain dataset
- Following LLaVA's architecture & training recipe



Baseline model – Baby model

Input



Where is the knife?

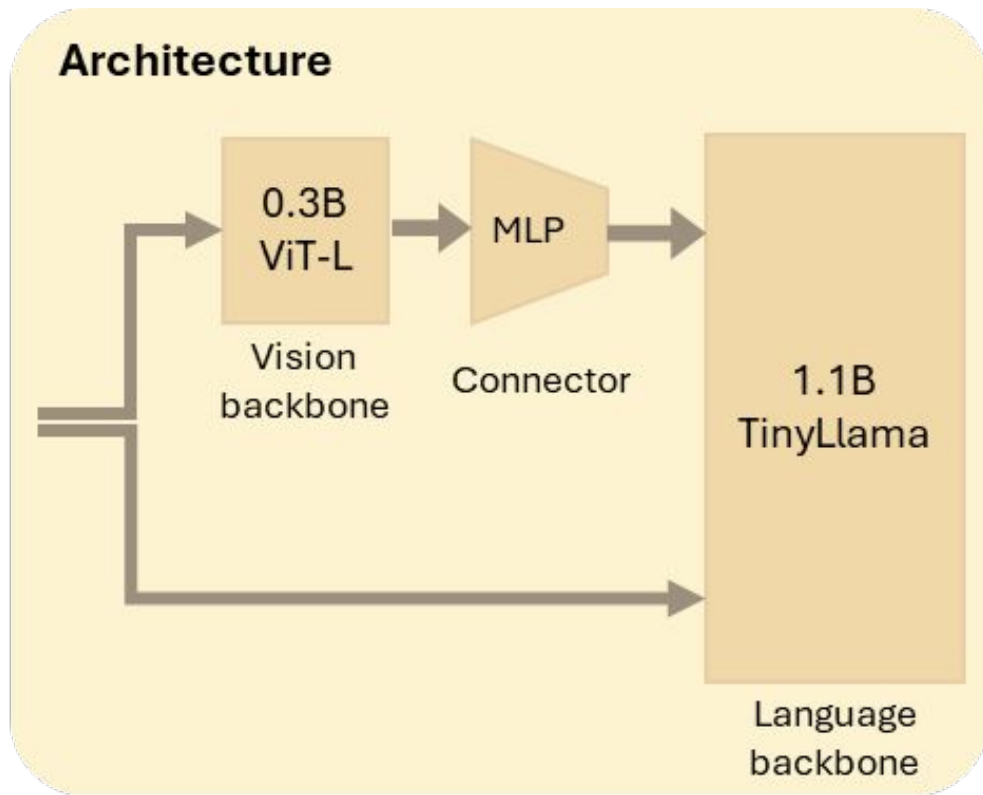


Which is a stroller?



Where does the toy leave the frame from?

Baseline model – Baby model



**Approx. parameters
(one total per model)**

Baby model
≈1.4B total

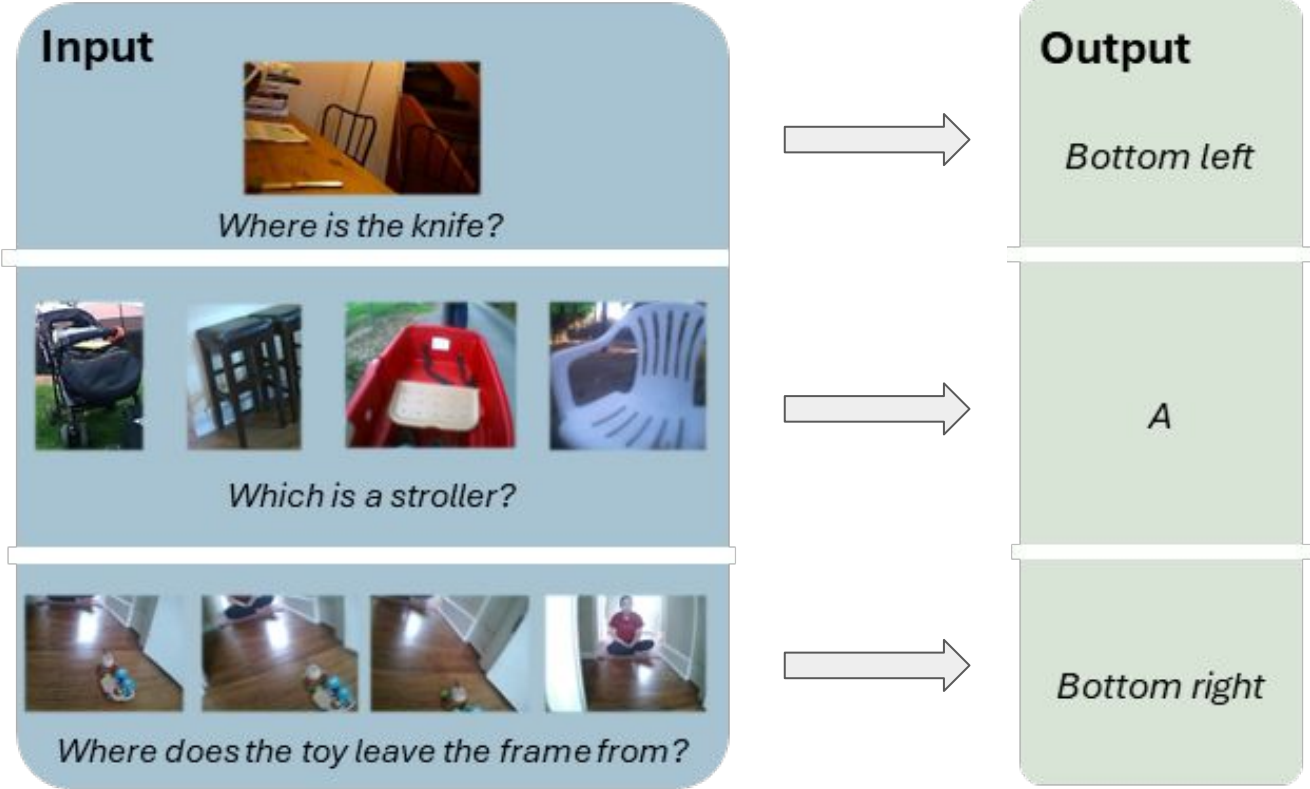
LLaVA-OneVision-7B
≈7B total

Qwen2.5-VL-7B
≈7B total

B = billion parameters. Approximate totals.

Vision backbone recipe: Dosovitskiy, Alexey, et al. "An image is worth 16x16 words: Transformers for image recognition at scale." arXiv preprint arXiv:2010.11929 (2020).
Language backbone recipe: Zhang, Peiyuan, et al. "Tinyllama: An open-source small language model." arXiv preprint arXiv:2401.02385 (2024).

Baseline model – Baby model



Training steps for the baby model

Stage 0 - Language

Unimodal pretrain

Dataset:

- Language only (283k)

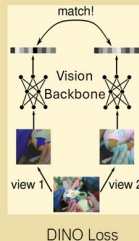


Stage 0 - Vision

Unimodal pretrain

Dataset:

- Vision only (1085k)



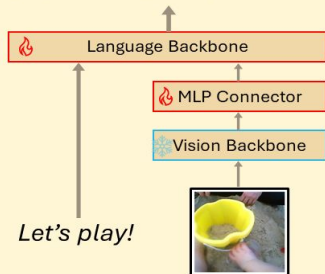
Stage 2

Joint pretrain

Dataset:

- Image-utterance (768k)
- Video-utterance (181k)
- Multi-turn (63k)

Let's play! can you put sand in your cup? and sand in the bucket?



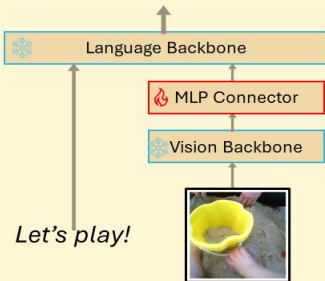
Stage 1

Feature alignment

Dataset:

- Image-utterance (768k)

Let's play! can you put sand in your cup? and sand in the bucket?



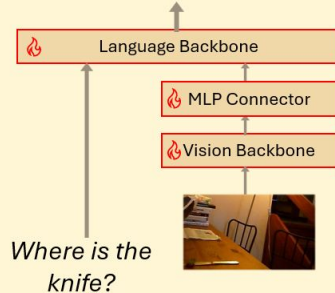
Stage 3

Instruction finetune

Dataset:

- Instruction set (113k)

Bottom left.



Training steps for the baby model

Stage 0 - Language

Unimodal pretrain

Dataset:

- Language only (283k)

*Let's play! can you put sand
in your cup? and sand in the
bucket?*

Language Backbone

Let's play!

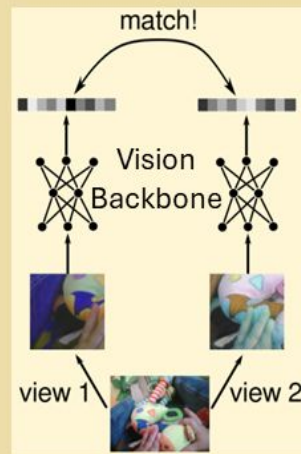
Autoregressive Loss

Stage 0 - Vision

Unimodal pretrain

Dataset:

- Vision only (1085k)



DINO Loss

Training steps for the baby model

Stage 0 - Language

Unimodal pretrain

Dataset:

- Language only (283k)

*Let's play! can you put sand
in your cup? and sand in the
bucket?*

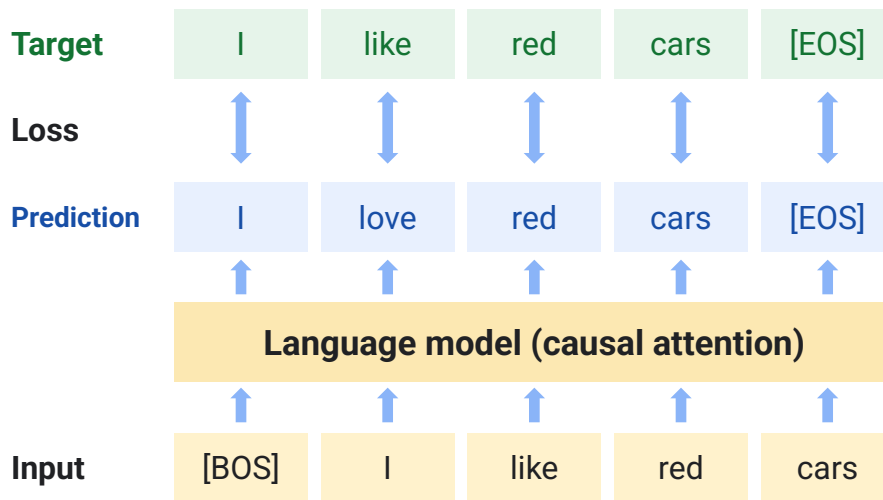
Language Backbone

Let's play!

Autoregressive Loss

Learn to predict the next token

Ground truth: input shifted one position left



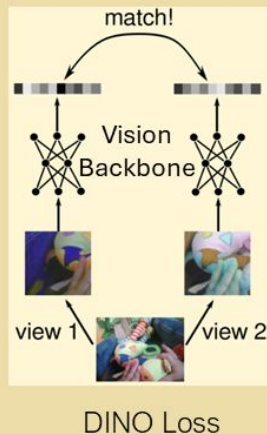
Training steps for the baby model

Stage 0 - Vision

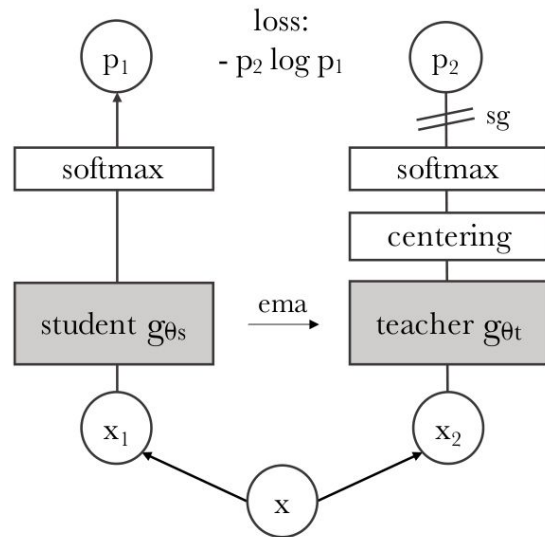
Unimodal pretrain

Dataset:

- Vision only (1085k)



DINO: match two views of one image



Train the student to match the teacher.
The targets come from another view—not human labels.

Training steps for the baby model

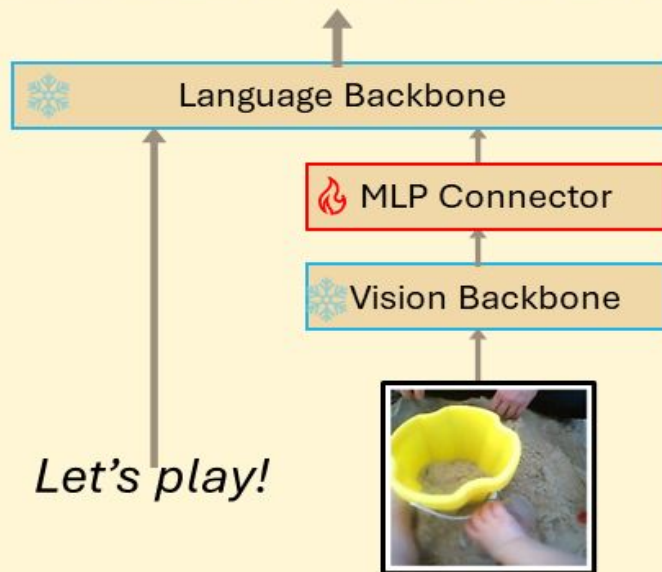
Stage 1

Feature alignment

Dataset:

- Image-utterance (768k)

Let's play! can you put sand in your cup? and sand in the bucket?



Training steps for the baby model

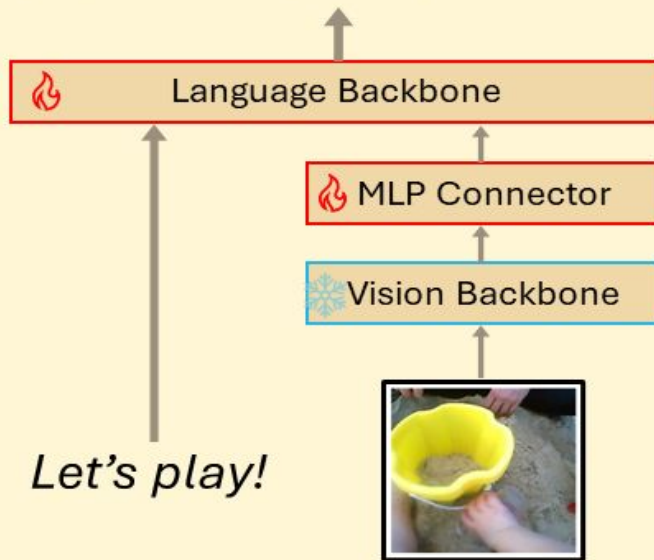
Stage 2

Joint pretrain

Dataset:

- Image-utterance (768k)
- Video-utterance (181k)
- Multi-turn (63k)

Let's play! can you put sand in your cup? and sand in the bucket?



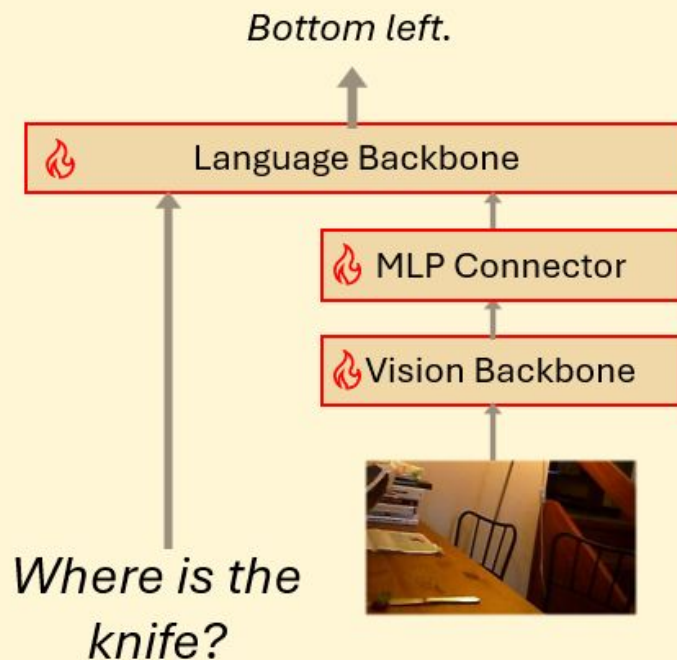
Training steps for the baby model

Stage 3

Instruction finetune

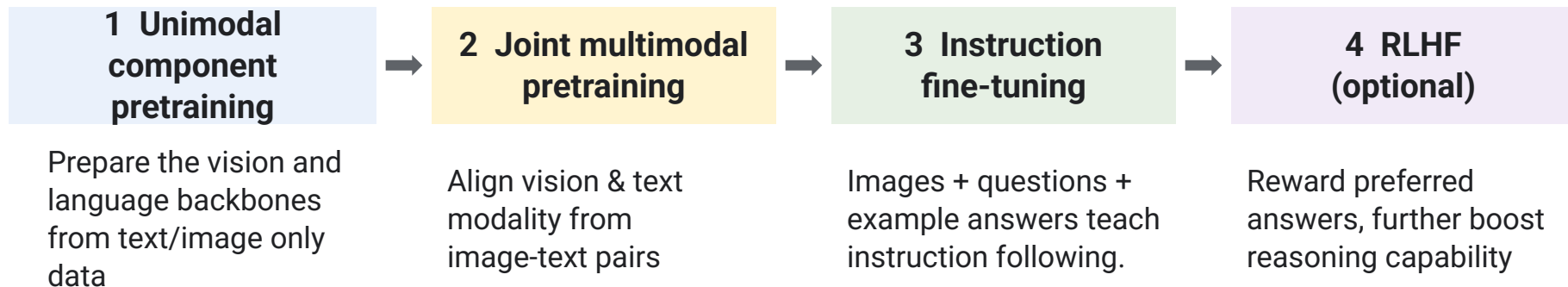
Dataset:

- Instruction set (113k)



How are vision-language models trained?

VLMs first learn initial data pattern from broad data, then learn to follow instructions.



BabyVLM-v2: Developmentally Plausible Benchmarks



BabyVLM-V2

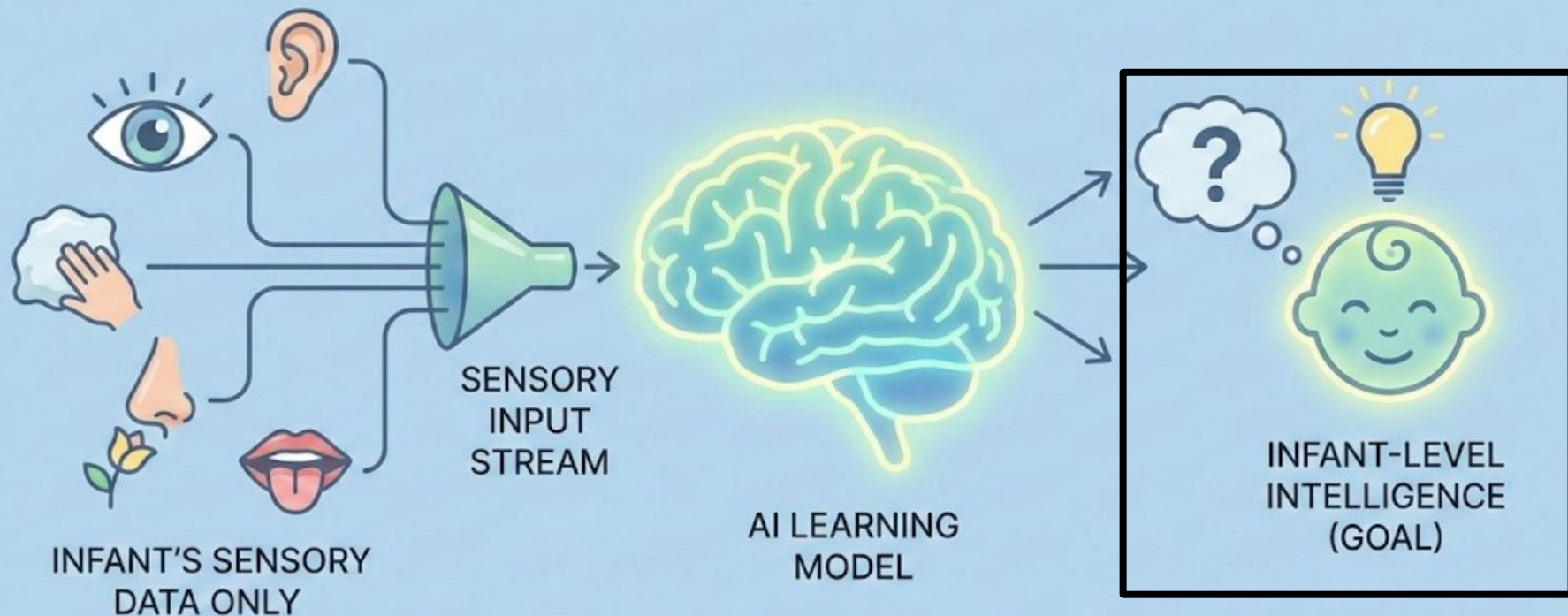
Max Whitton

maxwh@bu.edu



Can we achieve infant-level intelligence using infants' sensory data?

How to assess “infant-level intelligence”?





Can we achieve infant-level intelligence using infants' sensory data?

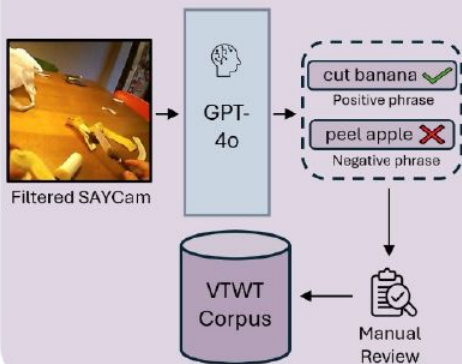
Developmentally **inspired** benchmarks [\[ICCV 2025\]](#) : *Intuitive, heuristic*

Labeled-S

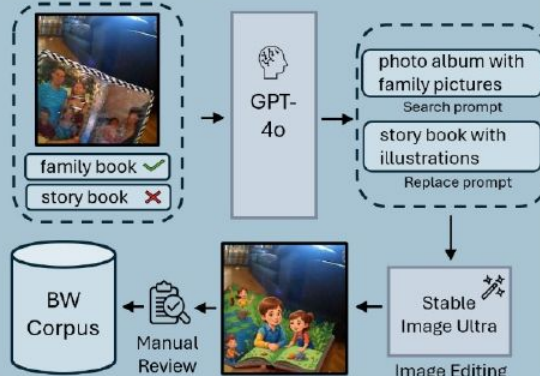


puzzle

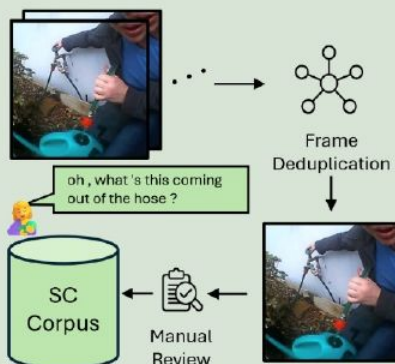
Visual Two-Word Test



Baby Winoground



SAYCam Caption





Can we achieve **infant-level intelligence** using infants' sensory data?

Developmentally **inspired** benchmarks [ICCV 2025] : *Intuitive, heuristic*

Developmentally **aligned** benchmarks

POSTER

📅 Sat, Jun 6, 2026 · 3:45 PM – 5:45 PM PDT

📍 ExHall A & F 211



BabyVLM-V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models

Shengao Wang · Wenqi Wang · Zecheng Wang · Max Whitton · Michael Wakeham · Arjun Chandra · Joey Huang · Pengyue Zhu · Helen Chen · David Li · Jeffrey Li · Shawn Li · Andrew Zagula · Amy Zhao · Andrew Zhu · Sayaka Nakamura · Yuki Yamamoto · Jerry Yokono · Aaron Mueller · Bryan A. Plummer · Kate Saenko · Venkatesh Saligrama · Boqing Gong





Can we achieve **infant-level intelligence** using infants' sensory data?

Developmentally **inspired** benchmarks [ICCV 2025] : *Intuitive, heuristic*

Developmentally **aligned** benchmarks

JAN 2025

How to assess
"infant-level
intelligence"?

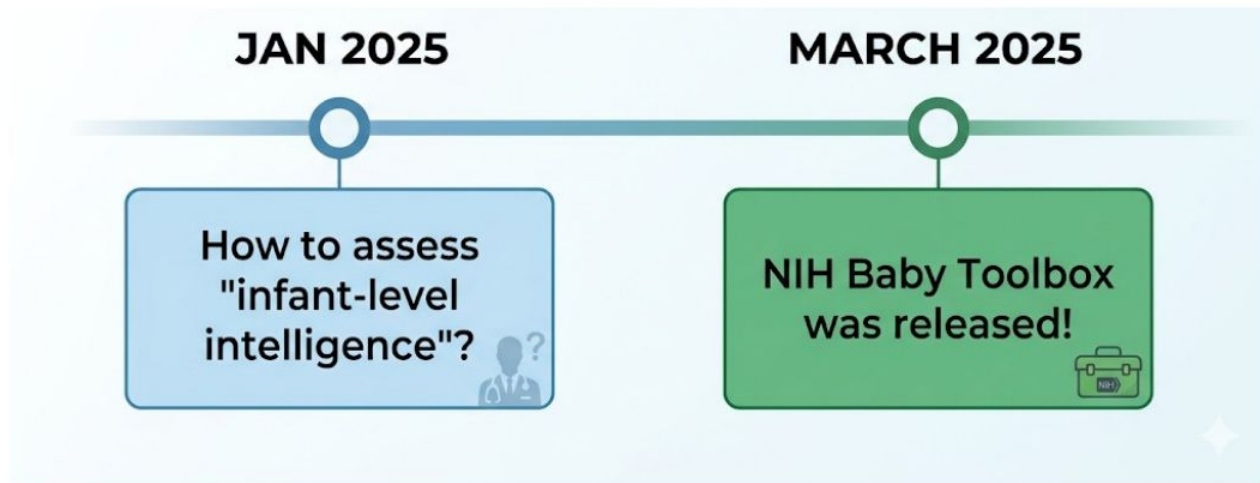




Can we achieve infant-level intelligence using infants' sensory data?

Developmentally **inspired** benchmarks [ICCV 2025] : *Intuitive, heuristic*

Developmentally **aligned** benchmarks



What is the NIH Baby Toolbox?



Developmental assessment through tablet games



1–42

months of age

31

tests

Cognition
Motor
Social-emotional

Tests matched to developmental age

Who uses the NIH Baby Toolbox?



Educators

School readiness



Clinicians

Developmental
assessment



Research

Developmental
intelligence

Visual delayed response



Does the baby understand
object permanence?

1.5x speed

Can they track moving objects?

Visual delayed response extended, domain-aligned

book



crib



pillow



book



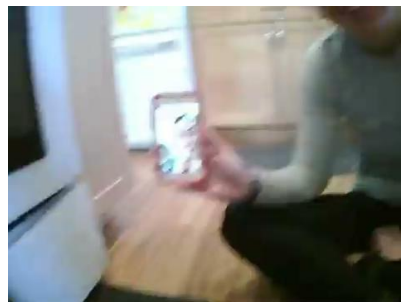
doll stuffed toy



book



phone



doll stuffed toy



DevCV Toolbox



Visual Delayed Response

"Where does the toy leave the frame from? Answer with top or bottom."

Answer: bottom



Memory

Round 1



Rounds 2-15: "Which is the new image? Answer with A or B."

Answer: A



Who Has More

"Which has more cups? Answer with A or B."

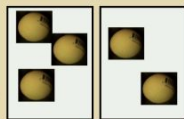
Answer: B



Who Has More

"Which has more balls? Answer A or B."

Answer: A



Subitizing

"How many chairs do you see? Answer with a number 1-12."

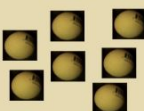
Answer: 3



Object Counting

"How many balls do you see? Answer with a number 1-12."

Answer: 7



NIH Baby Toolbox



Visual Delayed Response

"Look at that."
"Where is it hiding?"

Answer: *touch*



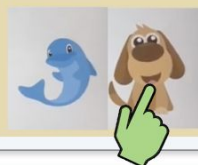
Memory

"The animal is hungry. Touch the animal!"



"Now you see two animals. One of them we already fed. Let's touch the new animal!"

Answer: *touch*



Who Has More?

"Who has more eggs?"

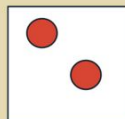
Answer: *touch*



Subitizing



1 second



"How many did you see?"

Answer: 2

Object Counting

"Count the blocks."

Answer: 4



DevCV Toolbox



Picture Vocabulary

"Which is a stroller?
Answer with A, B, C, or D."

Answer: A



Looking While Listening

"Which is a chair?
Answer A or B."

Answer: B



Localization

"Where is the knife?
Answer with top left,
top right, bottom left,
or bottom right."



bottom left

Left / Right



"Look at this cat. Which is exactly
like it? Answer with A, B, or C."

Answer: B



Spatial Details



"Look at this hand. Which
has a hand exactly like it?
Answer with A, B, or C."

Answer: B



NIH Baby Toolbox



Picture Vocabulary

"Touch the picture of chair."

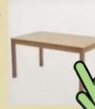
Answer: *touch*



Looking While Listening

"Touch the picture of
of table."

Answer: *touch*



Mullen RL #19

"Look at these.
Where is the cat?"

Answer: *point*



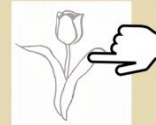
Mullen VR # 29

"Look at this boy. Find another
boy that is just like this one."

Answer: *point*



Mullen VR # 25



"Look at this flower. Look for one
just like it. Look for it here."

Answer: *point*



Memory



Can the baby remember objects, without them being named?

Memory

"The animal
is hungry.
Touch the
animal!"



"Now you see two animals.
One of them we already fed.
Let's touch the new animal!"

Answer: *touch*



Memory



Memory

Round 1



Rounds 2-15: "Which is the new image? Answer with A or B."

Answer: A



Who Has More?



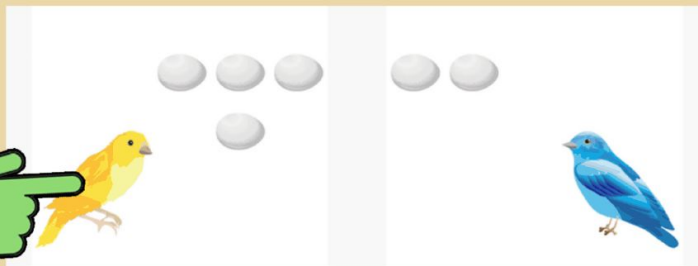
Can the baby compare two quantities of the same object?

Being able to count is not required!

Who Has More?

"Who has more eggs?"

Answer: *touch*



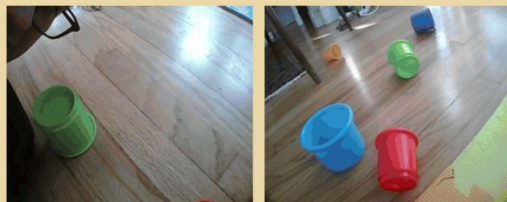
Who Has More?



Who Has More

"Which has more cups? Answer with A or B."

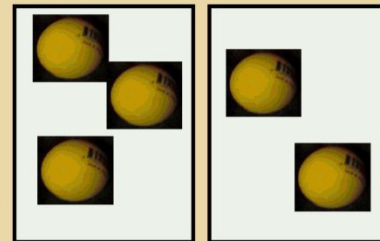
Answer: B



Who Has More

"Which has more balls? Answer A or B."

Answer: A



Object Counting



Can the baby count to 12?

The toolbox offers partial credit for counting consecutively up to a point, which we plan to implement for future work.

Object Counting

"Count the
blocks."

Answer: 4



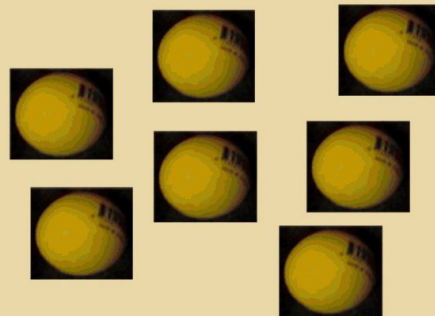
Object Counting



Object Counting

"How many balls do you see? Answer with a number 1-12."

Answer: 7



Picture Vocabulary



Can the baby recognize objects by name?

Extra challenge: the distractors are semantically related!

Picture Vocabulary

"Touch the picture of chair."

Answer: *touch*



Picture Vocabulary



Picture Vocabulary

“Which is a stroller?
Answer with A, B, C, or D.”

Answer: A



Localization



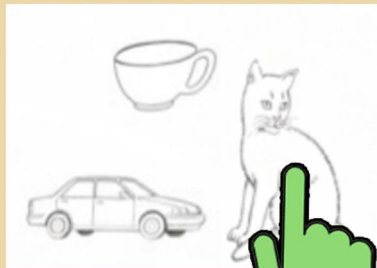
Can the baby point at a named object?

Extra challenge: distractor objects

Mullen RL #19

"Look at these.
Where is the cat?"

Answer: *point*



Localization



Localization

“Where is the knife?
Answer with top left,
top right, bottom left,
or bottom right.”



bottom left

Mullen Visual Reasoning #29

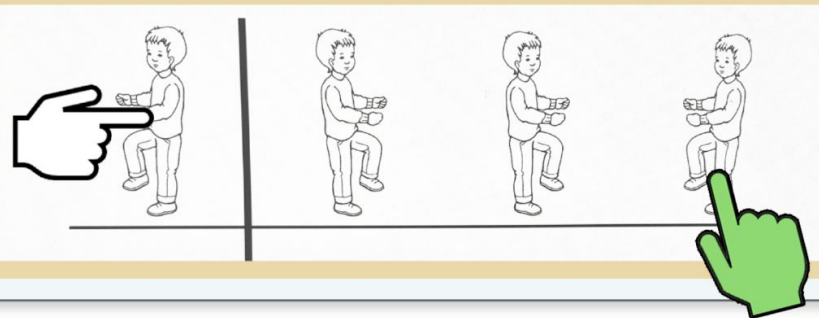


Can the baby match objects by orientation?

Mullen VR # 29

"Look at this boy. Find another boy that is just like this one."

Answer: *point*



Left/Right



Left / Right



"Look at this cat. Which is exactly like it? Answer with A, B, or C."

Answer: B



Mullen Visual Reasoning #25



Can the baby remember the details of a particular instance of an object versus others from its category?

Mullen VR # 25



"Look at this flower. Look for one just like it. Look for it here."

Answer: *point*



Spatial Details



Spatial Details



“Look at this hand. Which
has a hand exactly like it?
Answer with A, B, or C.”

Answer: B





Can we achieve **infant-level intelligence** using infants' sensory data?

Developmentally **aligned** benchmarks

POSTER

📅 Sat, Jun 6, 2026 · 3:45 PM – 5:45 PM PDT

📍 ExHall A & F 211



BabyVLM-V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models

Shengao Wang · Wenqi Wang · Zecheng Wang · Max Whitton · Michael Wakeham · Arjun Chandra · Joey Huang · Pengyue Zhu · Helen Chen · David Li · Jeffrey Li · Shawn Li · Andrew Zagula · Amy Zhao · Andrew Zhu · Sayaka Nakamura · Yuki Yamamoto · Jerry Yokono · Aaron Mueller · Bryan A. Plummer · Kate Saenko · Venkatesh Saligrama · Boqing Gong



BabyVLM-v2: Instruction Tuning

Toward **Developmentally Plausible** Vision Foundation Models

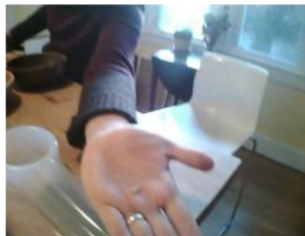


BabyVLM-V2

Wenqi Wang

wqwang@bu.edu

From infant experience to model-ready data

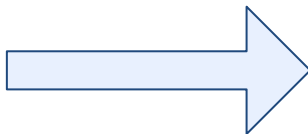


Recorded experience

3 infants

478 hours

~6–32 months



Model-ready data

Pretraining

Instruction
Tuning

Benchmark

What is instruction tuning?

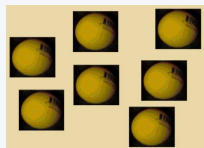
Instruction tuning

Use instructions + target responses to teach instruction following.

SFT: supervised fine-tuning

Update a pretrained model using labeled input–target examples.

In BabyVLM-v2, instruction tuning is done with SFT.



“How many balls?”

Image + instruction

+

Target

7



Update weights

Learn the target response

Training: target provided. Evaluation: answer hidden.

Why do we need instruction tuning?

In Pretraining



“Can you do
the white one?”

Naturalistic language—not explicit task

In benchmarking



How many balls?
Answer with 1–12.

Target: 7

Visual input + instruction + target

Mismatch!

SFT updates a pretrained model using labeled input–target examples.

Here, instruction tuning teaches task instructions and answer formats.

Instruction data - annotations become task supervision

1 Retrieve evidence

Labels, regions, tracks

2 Construct the task

Choices, crops, or sequences

3 Create supervision

Input + instruction $\rightarrow$ target

Language grounding

Labels + crops + box centers

**Picture Vocabulary · Looking While
Listening · Localization**

Visual matching

Crops + orientation + context

Left/Right · Spatial Details

Temporal / memory

Object events or trial history

Visual Delayed Response · Memory

Quantity

Natural counts or repeated
crops

**Who Has More · Subitizing
Object Counting**

Single image, multiple images, or multiple turns — one text interface.

Instruction data - annotations become task supervision

Example: *picture vocabulary* task construction

Picture Vocabulary

"Which is a stroller?
Answer with A, B, C, or D."

Answer: A



Frames sampled from
SAYCam

We develop data samples constructor for each task in DevCV Toolbox

Instruction data - 10 tasks, 2 held out w./o SFT data

≈113K

Supervised Finetuning

≈13.6K

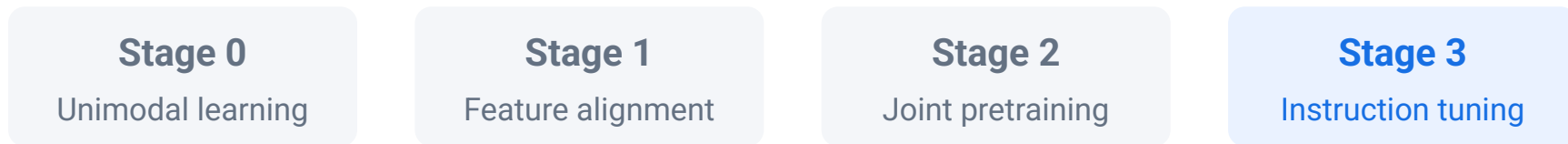
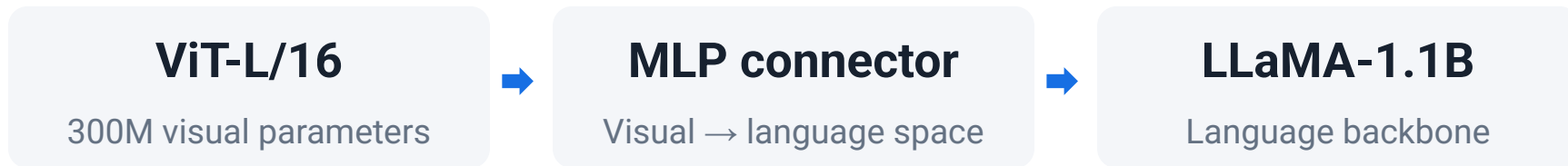
Validation

≈17.8K

Test (DevCV benchmark)

Task	SFT (K)	Validation (K)	Test (K)
Picture Vocabulary	22.4	0.7	1.1
Looking While Listening	0	0	1.2
Localization	12.3	2.1	2.1
Left/Right	12.3	2.2	2.3
Spatial Details	11.8	1.2	1.2
Visual Delayed Response	2.6	0.4	0.5
Memory	12.0	0.3	0.3
Who Has More (synthetic / natural)	11.2 / 12.4	1.8 / 1.9	1.8 / 2.2
Subitizing (synthetic / natural)	0 / 0	0 / 0	1.9 / 0.2
Object Counting	13.7	3.0	3.0

Setup - A compact model, with SFT as the final stage



Stage 3 updates all modules with an autoregressive objective.

5 epochs

Global batch: 128

$1e-4$ / $5e-4$

Vision / connector + language

≈24 hours

4 × A6000, 48 GB each

BabyVLM-v2: Evaluation & Experiments

Toward **Developmentally Plausible** Vision Foundation Models



BabyVLM-V2

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Evaluation — In-domain and two domain shifts

Baby model: SAYCam pretraining + mixed instruction tuning

SAYCam(in-domain)



In-domain

Infant-egocentric imagery
Held-out source split
Adapted DevCV tasks

Ego4D(out of domain)



New naturalistic domain

Adult-egocentric imagery
Different environments
Same task construction

Original NIH Toolbox (out of domain)



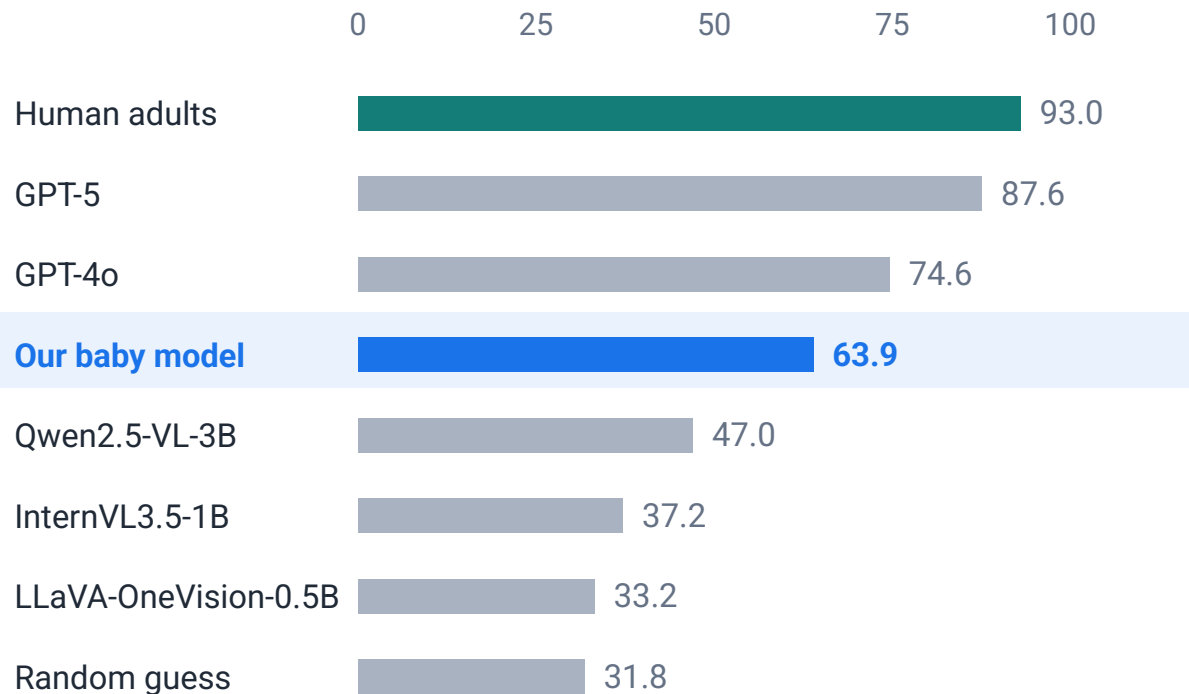
Original stimuli

Cartoon / abstract imagery
Small task-specific sets
Report correct / total

Held out from SFT: Looking While Listening and Subitizing.

Main result — 63.9% on in-domain DevCV

Selected reference models and the adult survey



63.9%

Overall accuracy

Above the compact checkpoints shown.

Not matched for data or adaptation.

Main result — An uneven skill profile

Task accuracy (%) · gap = baby model minus GPT-4o

Task	Baby model	GPT-4o	Gap (pp)
Object Counting	47.3	39.0	+8.3
Left/Right	96.4	89.8	+6.6
Spatial Details	92.8	92.6	+0.2
Who Has More — synthetic	99.7	87.9	+11.8
Picture Vocabulary	32.4	93.7	-61.3
Localization	37.8	81.7	-43.9
Who Has More — naturalistic	60.5	79.3	-18.8

Controlled synthetic comparisons do not guarantee success in natural scenes.

Validation – Instruction data helps other VLMs

A separate validation experiment: one task-specific model per task.

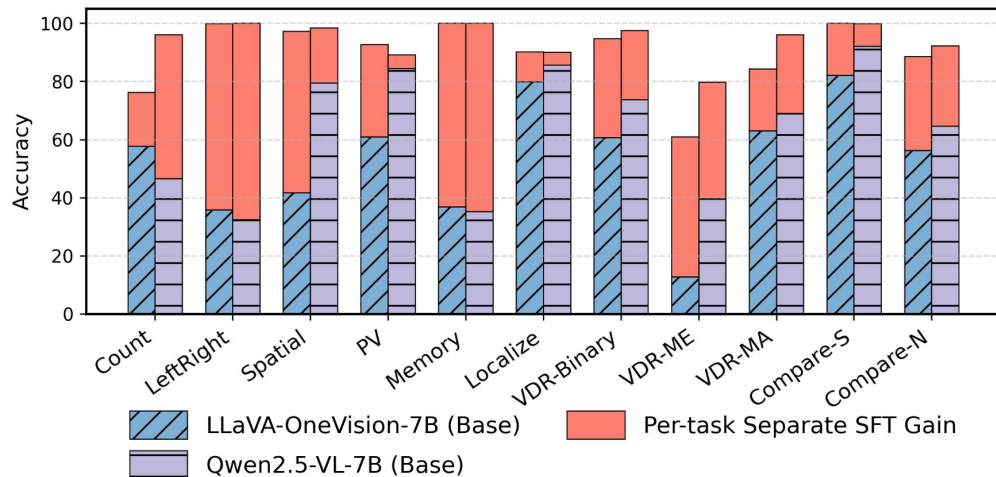
LLaVA-OneVision-7B/
Qwen2.5-VL-7B



Task-specific
LoRA fine-tuning



Gains across
tasks

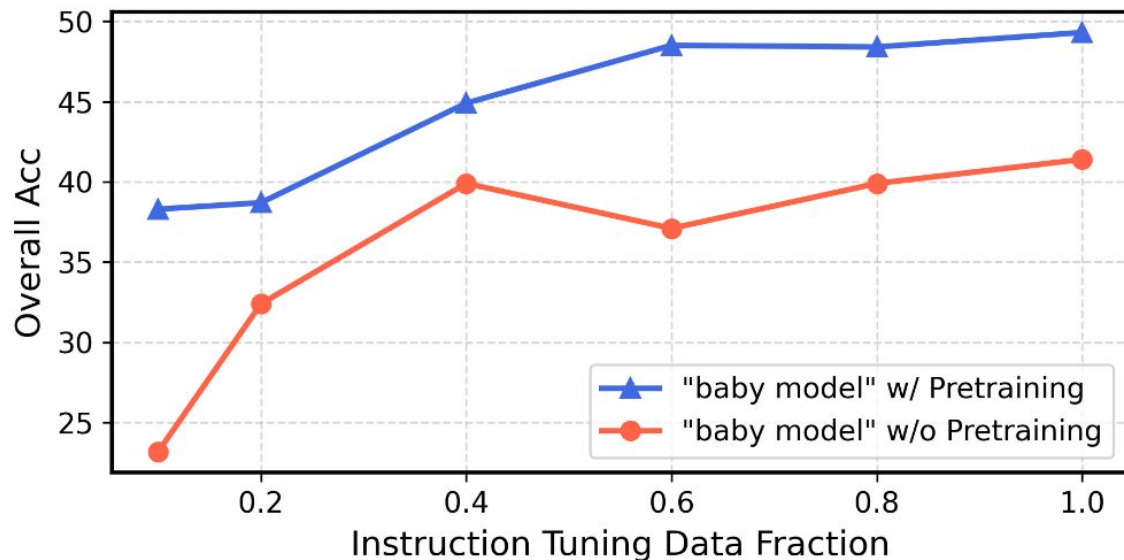


5 epochs · batch 128 ·
learning rate 1e-4

Useful supervision—effectively
guides models toward the
tasks in DevCV Toolbox.

Ablation — Pretraining helps data efficiency

Same architecture · vary the instruction-data fraction · evaluate after one epoch



**Pretraining wins
at every fraction.**

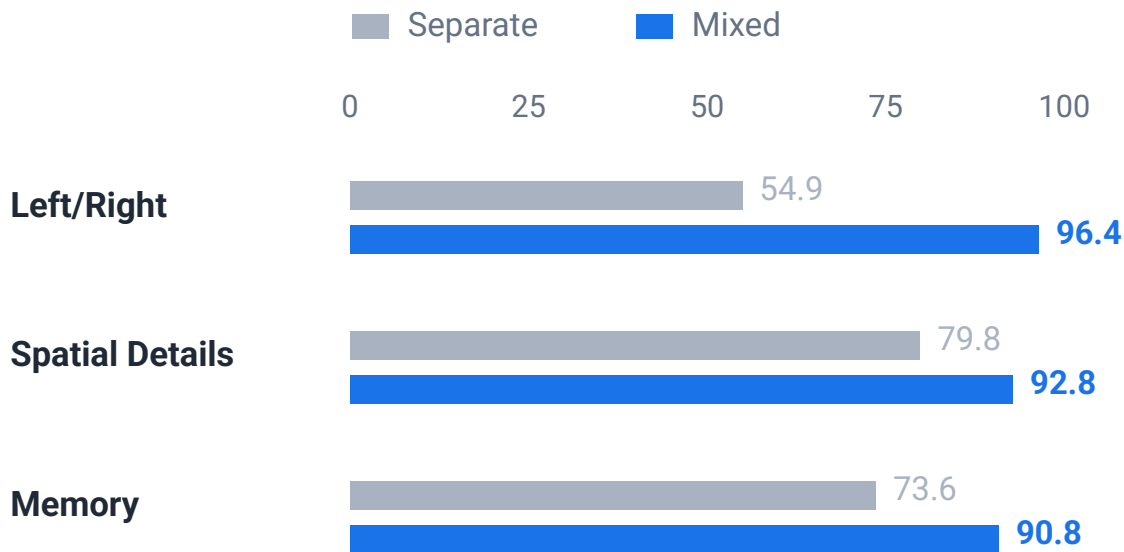
Largest advantage
when instruction
data is limited.

Stages 0–2 + SFT
vs. SFT from scratch

Task supervision complements earlier experience; it does not replace it.

Ablation — Mixed-task SFT improves the overall score

Separate task-specific models vs. one jointly tuned model



Overall accuracy

56.9 → 63.9

+7.0 points

One model,
multiple task formats.

Not every metric improves: naturalistic Who Has More, 62.1 → 60.5.

Ablation — Natural caregiver speech remains competitive

Change the pretraining language; keep the architecture and downstream SFT fixed.

Overall DevCV accuracy (%)

Caregiver utterances  **63.9**

GPT-4o captions  **65.4**

+1.5 points

Only a modest overall gain.



GPT-4o: Look at the cute chalk drawing!

Original: No I drew his coat on so I protected him from the mud.

Ablation — Filtering quality matters

Same model and downstream SFT; different pretraining filters

Overall DevCV accuracy (%)



67.3%

Strict filtering

Less than half
the default
pretraining data.

Better retained pairs—not simply more pretraining pairs.

Limitation — Transfer across visual domains is limited

The same task builders, but different visual experience

SAYCam → Ego4D

Metric	SAYCam	Ego4D	Random
Overall	63.9	48.8	31.8
Spatial Details	92.8	47.1	33.3
Memory	90.8	72.4	25.0

Original NIH Toolbox

Correct / total · small sets

Who Has More	13 / 24
Object Counting	2 / 6
Mullen Visual Reception	3 / 12

Above chance on Ego4D, but the in-domain skill profile does not fully transfer.

Limitation — Related tasks do not yet transfer

Two task families receive no instruction-tuning examples.

Seen during SFT

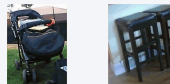
Held out from SFT

Outcome

Picture Vocabulary



Looking While Listening



4 → 2 choices

Near chance

50% random

Object Counting



1 image
1–12 objects



Subitizing

Blank

1–4

Blank

Near chance

25% random

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Q&A

Evaluation & Experiments

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BabyVLM-V2

BabyVLM-v2: Future Work

Toward **Developmentally Plausible** Vision Foundation Models



BabyVLM-V2

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Is touch a mechanism for human learning?

Making Sense of Touch from the Child's View for Contrastive Learning

Max Whitton, Zecheng Wang*, Puchen Liu*, Quang Tuan Truong, Shengao Wang, Manaswi Yadamreddy, Oktay Ozel, Visista Jayanti, Saniya Sekhon, Hanna Samuel Tadesse, Lawrence Miao, Junjie Wang, Jiasen Lu, Chen Yu¹, and Boqing Gong

Project Homepage: https://max-whitton.github.io/Contrastive_Learning_From_Touch

Abstract—Is the sense of touch a mechanism for human babies' learning of visual concepts? If so, can we quantify its importance, and to what extent do babies rely on their sense of touch for visual learning? To approach these questions in a principled way, we propose a structured coding system for baby-centric touch events, yielding a dataset of 264k two-second clips of touch events coded according to this system. Using this dataset, we pretrain developmentally grounded models that reveal promising insights into the nature of baby learning from touch.

input as we can from their real life, without adding any extra data that a baby would not have access to. However, it is prohibitively difficult to capture a baby's high-fidelity tactile sensations at scale directly. We approach this problem in a principled way, without introducing intrusive physical sensors, by leveraging a structured coding system for babies' touch events. We apply this taxonomy of touch to longitudinal, egocentric video recordings from three infants'

Oral Session 1 for more information!

Jun 2026

Announcement: The BabyVLM Challenge!



BabyVLM Challenge

Pretraining vision-language models on developmentally grounded data
inspired by infant cognition and learning.

[Submit Results](#)[Explore Pretraining](#)[Explore Task](#)

ABOUT

Overview

The BabyVLM Challenge is a shared task run in partnership with the [BabyLM Challenge](#), focused on developmentally plausible, sample-efficient vision-language modeling. Participants work with new datasets and evaluations purpose-built for small vision-language models trained on limited data.

The challenge will be introduced at the [BabyVLM workshop](#) with a tutorial. No prior BabyLM experience is needed to participate.



Pretraining

Pretrain on a compact dataset of egocentric audiovisual streams, provided below.



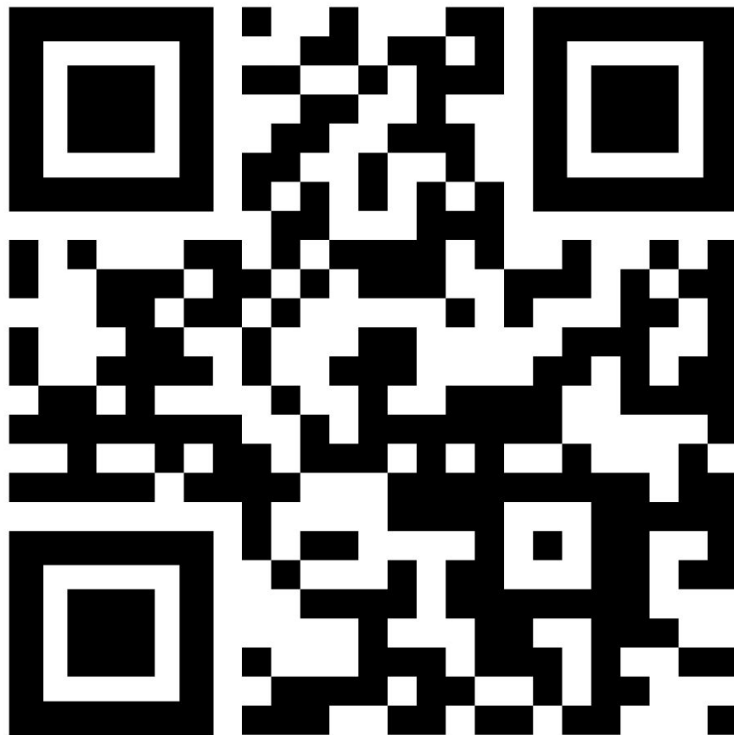
Task

Report scores on the 10 subtasks of the DevCV Toolbox.



Deadline

Coming soon!



Q&A



BabyVLM-V2

Until 12:50